

The *curated* short-stay marketplace for the GCC.

A tech marketplace operated by Keys Please Holiday Homes Rental Dubai LLC — 23 years of property operator expertise applied to curation and trust, not platform features. v8.2 locks three principles: (1) all-inclusive rates (utilities are a host operating cost), (2) authoritative DET TDF matrix (admin-editable, AED 10–100/night, B.50 TDFRateSchedule), (3) **5% VAT applies uniformly** across all guest-charged line items — accommodation (regardless of stay length), cleaning, Experiences, Services, DSS commission. Only TDF is not vatable. "Hotel Apartment" renamed to "SR" (Serviced Residence). The previous "Homes = Exempt under Article 45" treatment was a misreading of law and is corrected. 45 entities locked. Engineering reads this file.

VERSION	DATE	OPERATING ENTITY	DET LICENCE
v8.4 canonical <i>admin-gated</i> <i>channel distribution</i>	23 May 2026	Keys Please Holiday Homes Rental Dubai LLC	#727937 · since 2003
STATUS	ENTITY COUNT		
Development-ready · 14 ADRs locked · tech marketplace	44 locked (28 commerce + 16 platform-ops) v8.4: + B.52 <i>ChannelDistribution</i>		
HOMES COMMISSION	EXPERIENCES	SERVICES	SERVICE SLA
12%	20%	25%	2 hours
PAYOUTS	TOURISM DIRHAM		
D+1	10–100 AED		

0.1 Abstract

DubaiShortStay.com is a **tech marketplace** for curated short-stay accommodation in the UAE/GCC, operated by **Keys Please Holiday Homes Rental Dubai LLC**, a property management company in business since 2003 under DET Licence #727937. Keys Please's 23 years of operator expertise inform platform's curation discipline, host onboarding playbooks, and customer-experience standards — but DSS—the-platform does not perform property operations. Hosts (registered holiday-homes operators) hold their own DET permits and operate their listed properties. DSS mediates discovery, bookings, payments, communications, disputes, insurance coverage, tax remittance, and OTA distribution.

The platform serves three product lines (Homes, Experiences, Services) operated as a single marketplace by Keys Please Holiday Homes Rental Dubai. *Multi-tenant SaaS ambitions were dropped in v8.3* — hosts are accounts within the single platform, not separate tenants. Future Spain / GCC expansion (Phase 4+) would be a separate deployment, not a tenant slice.

This document is the master consolidated Software Requirements Specification covering versions v6.1 through v7.5. v6.x defined the consumer entity and locked the brand, commercial rules, and page architecture. v6.4 added the transactional layer; v6.4.1 closed transactional edge cases. v7.0 introduced the enterprise architecture; v7.0.1 and v7.0.2 closed remaining entity gaps. v7.1 added Insurance/Claim (provisional values pending business confirmation). v7.2 and v7.3 are outlines for Phase 5+ native mobile and sovereign cloud variants. v7.4 added long-stay workflows. **v7.5 locked the marketplace positioning** — decommissioned operator-side entities (HousekeepingTask, MaintenanceTicket, Vendor) that had crept in; reframed InspectionReport DepositInspection (host+guest joint, marketplace-neutral); reframed RelocationPackage as marketplace Service bundle.

33 entities are locked. 14 architecture decision records are locked. Engineering may begin development. The phased roadmap in §17 describes the path from prototype through Phase 5 platform.

0.2 Scope & status

What this document covers

- The consumer marketplace surface (web + mobile-responsive)
- 28 locked commerce entities — Property, Experience, Service, Provider, Owner, User, Booking, Payment, Refund, Review, BookingModification, DisputeCase, PayoutBatch, Wishlist, PromoCode, RatePlan, PricingRule, EmailTemplate, CorporateAccount, InsurancePolicy, Claim (B.33–B.36 long-entities decommissioned v8.3). **UtilityCharge decommissioned v8.2 — rates are all-inclusive. See §B.32.** + 16 platform-ops entities (B.37 BlogPost through B.52 ChannelDistribution). 4 long-stay entities (B.33–B.36) decommissioned in v8.3 — see tombstone.
- The commercial model (commission tiers, payout schedule, cancellation policies, tax handling, TDF collection)
- The enterprise platform target architecture (Channel Hub at Phase 3, multi-tenancy)
- OTA integration via Channel Manager (Channex/Hostaway per ADR-001) — Airbnb, Booking.com, Vrbo, Expedia, Agoda
- Compliance regime (UAE PDPL, DET, FTA VAT/TDF, PCI DSS, GDPR for EU expansion)
- Claim lifecycle (provisional cover values)
- **Long-stay scope** — DECOMMISSIONED v8.3. DSS is a short-stay marketplace. See §B.33–B.36 decommission tombstone. Rates remain all-inclusive; utilities are a host operating cost, not a guest or DSS concern (v8.2)
- Phase 5+ outlines — native mobile (v7.2) and sovereign cloud (v7.3)

What remains pending

- **E1 strategic decision** — DSS Coverage policy specifics (cover amount, underwriter, exclusions) — schema is locked in v7.1, business confirmation needed before public launch
- Detailed mobile component spec — activate on Phase 5 trigger
- Detailed sovereign deployment spec — activate on government deal trigger

⚠ **E1 status:** moved from BLOCKED to PROVISIONAL in v7.1. Engineering can build the Insurance + Claim entities with industry-standard defaults. Business confirms final values before public launch; final values swap via config without schema changes.

0.3 Glossary

Terms used throughout the SRS. Alphabetized. Includes UAE-specific regulatory terminology, hospitality industry acronyms, and DSS-internal vocabulary.

A–C

TERM	DEFINITION
ADR	Architecture Decision Record. See <code>/docs/adr/</code> in the engineering repository. 14 ADRs locked.
ADHA	Abu Dhabi Health Authority (sector-specific compliance, sovereign cloud variant v7.3).
AML	Anti-Money Laundering. Relevant to KYC L1/L2 escalation and payment provider onboarding.
Anchor host	Keys Please as the launch host on DSS with ~200 properties. Not an entity field; an internal designation per v7.5 §1.
BNPL	Buy Now Pay Later. Tabby and Tamara provide this in UAE/KSA.
Booking	The core commerce entity, §B.9. Lifecycle defined in §O. 12 states per §C.4.
CBUAE	Central Bank of UAE. Regulator for payment providers and 3DS/SCA requirements.
Channel Manager	External vendor (Channex, Hostaway) that manages OTA protocol details. ADR-001 locks integrate-not-build.
Chargeback	Payment reversal initiated by guest's bank via Visa/Mastercard dispute. v6.4.1 §Q.5.
Coverage	DSS Coverage — proprietary insurance product layered on every booking. v7.1 §B.30. Currently PROVISIONAL.
CSA STAR	Cloud Security Alliance Security, Trust & Assurance Registry. Certification track for sovereign cloud (v7.3).

Curation tier

top

dubai_favorite

top_provider

Unified across Stays, Experiences, Services. §B.0–B.2.

D–F

TERM	DEFINITION
D+1 payout	Owner receives funds one business day after guest check-in. Batched daily per §B.17 PayoutBatch.
DepositInspection	Joint host+guest condition documentation for stays with SecurityDeposit. v7.5 renamed entity §B.36.
DESC	Dubai Electronic Security Centre. Required certification for Dubai government deals (sovereign v7.3).
DET	Dubai Department of Economy and Tourism. Issues holiday homes operator permits. Keys Please holds DET #727937.
DisputeCase	Entity tracking dispute resolution. §B.16. 4h SLA for critical disputes.
DSS	DubaiShortStay.com — the tech marketplace platform.
DTCM	Dubai Tourism Commerce Marketing (legacy name; now under DET). Relevant for Experience permits.
Ejari	Dubai's mandatory tenancy contract registration system. Required within 30 days for leases ≥ 90 nights. v7.4 §B.34.
extended_stay Decommissioned v8.3	Stay classification ≥ 90 nights — removed in v8.3. All bookings are short-stay. The stay_classification field is dropped from Booking entity.
FTA	UAE Federal Tax Authority. VAT registration and quarterly returns.

G–K

TERM	DEFINITION
G42	UAE government-backed cloud (Core42 Cloud). Sovereign cloud option per v7.3.
GDPR	EU General Data Protection Regulation. Relevant for Phase 4 Spain expansion.

GMV	Gross Merchandise Value. Total booking value before commission.
HMAC	Hash-based Message Authentication Code. Used for webhook signature verification (Stripe, Tabby, Channex).
Host	The legal entity that lists properties on DSS. Holds DET permit per property. Same as Owner in entity §B.12.
IRPF	Impuesto sobre la Renta de las Personas Físicas (Spanish income tax). Relevant for Phase 4 Spain.
MORCEL	Internal name for the AI suggestions panel surface in the admin dashboard. v7.0 §15.
KP	Keys Please Holiday Homes Rental Dubai LLC. The operating company behind DSS-the-platform.
KYC	Know Your Customer. DSS escalation model: L0 (email-verified) → L1 (phone + ID document) → L2 (Emirates ID + payment method). §B.13.
KYC L0/L1/L2	KYC verification levels. L0 = guest browsing; L1 = booking under AED 5,000; L2 = bookings above or trust-sensitive flows.

L–O

TERM	DEFINITION
LCP	Largest Contentful Paint. Web performance metric. DSS target: p95 < 1.5s for property pages.
ShortStayAgreement <i>Decommissioned v8.3</i>	Formerly the agreement instrument for extended_stay bookings. Removed in v8.3 — Booking record itself is the contract artefact for short stays. No separate agreement flow.
LOI	Letter of Intent. Non-binding commitment from underwriter. Required before v7.1.1 lock.
Makani	Dubai geo-location code. Optional Property field for precise addressing.
me-central-1	AWS region in UAE (Bahrain DC). DSS primary region per ADR-014.
monthly <i>Decommissioned v8.3</i>	Stay classification 28–89 nights — removed in v8.3. All bookings are short-stay.
MSK	Amazon Managed Streaming for Apache Kafka. ADR-005 lock for Phase 2+ event bus.
NESA	UAE National Electronic Security Authority. Information Assurance Standards required for federal contracts (sovereign v7.3).
OTA	Online Travel Agency. Airbnb, Booking.com, Vrbo, Expedia, Agoda.
Outbox pattern	Pattern preventing dual-write inconsistency between DB writes and event publishes. Pre-development audit F1.2.
OXI	Booking.com's API/integration certification programme.

P–R

TERM	DEFINITION
Pattern A / Pattern B (TDF)	TDF remittance model. Pattern A = host collects + remits. Pattern B (DSS choice per v7.5 §7) = DSS collects + remits on host's authorisation.
PayoutBatch	D+1 daily payout batch. \$B.17. Finance review gate before bank submission.
PCI DSS	Payment Card Industry Data Security Standard. DSS targets SAQ-A scope (no PAN on infrastructure).
PDPL	UAE Personal Data Protection Law. Federal Decree-Law No. 45 of 2021. Primary data residency + subject rights framework.
PMS	Property Management System. DSS is NOT a PMS — it's a marketplace (v7.5 §1).
POI	Point of Interest. Entity \$B.5 for landmarks, beaches, malls, neighborhoods relevant to a property.
PriceLabs	Third-party AI dynamic pricing product. ADR-013 locks for Phase 2.
PROVISIONAL	Marker on entities/fields whose schema is locked but specific values await business confirmation. Currently applies to E1 (DSS Coverage values).
RBAC	Role-Based Access Control. 13 platform roles + 3 host roles per §4 (simplified v8.3 — tenant-role concept removed alongside multi-tenancy).
RDC	Rental Disputes Centre (Dubai). Alternative dispute resolution venue for leases. v7.4 \$B.34.
RERA	Real Estate Regulatory Agency (Dubai). Property registration authority.
RLS	Row-Level Security (Postgres). Multi-tenancy isolation mechanism at MVP per ADR-004.
RTB	Request to Book. Booking that requires host approval (alternative to Instant Book). \$B.O.
RTO / RPO	Recovery Time Objective / Recovery Point Objective. DR posture: 1-hour RTO, 5-minute RPO. ADR-014.

S–Z

TERM	DEFINITION
SAQ-A	PCI Self-Assessment Questionnaire A. The lightest PCI scope; requires no PAN data touching DSS infrastructure.
SecurityDeposit	Guest-funded held amount for long-stays. \$B.33. Card hold ≤ AED 30K, escrow above.
short_term	Stay classification 1-6 nights. Default workflow per v7.4 §V.
SLA	Service Level Agreement. Critical: 2h response for Services, 4h for critical disputes, 14 business days for claim payout.
SOC 2 Type II	System and Organization Controls 2, Type II. Enterprise SaaS certification. Phase 3 posture work.
SQS	Amazon Simple Queue Service. MVP event transport per ADR-005.
SSO / OAuth / OIDC	Single Sign-On / OAuth 2.0 / OpenID Connect. Federation patterns. UAE Pass at Phase 4.
Stripe Elements	Embedded payment form. DSS uses Elements (not Hosted Checkout) for branded payment flow.
Tabby	UAE/KSA BNPL provider. 4-installment plans. ADR-006.
Tamara	BNPL provider for KSA/Egypt expansion. Phase 2 secondary per ADR-006.

TDF	Tourism Dirham Fee. Per-property-per-night government fee set by Dubai's Department of Economy & Tourism (DET). Rate band AED 10–100/night across the 4-class × 6-size matrix (§T.2.2). Capped at 30 consecutive nights per stay. Collected by DSS at confirmation, remitted DET monthly (Pattern B per v7.5 §7). Rates stored in admin-editable TDFRateSchedule (§B.50), not hardcoded.
Tech marketplace	DSS's regulatory category. Distinct from "operator of record" — hosts hold operator permits, not DSS. v7.5 §1.
Tenant <i>Decommissioned v8.3</i>	Previously referred to an isolated customer scope on the platform (multi-tenant SaaS model). Removed in v8.3. DSS is a single marketplace; hosts are accounts within it, not separate tenants. See §3 single-platform model.
TRN	Tax Registration Number (UAE FTA). Required for VAT-compliant corporate invoices. §B.29.
UAE Pass	UAE federal digital identity. Federation candidate for Phase 4 (sovereign cloud variant).
UtilityCharge <i>DECOMMISSIONED v8.2</i>	Removed v8.2. Utilities are not a DSS concept. Rates are all-inclusive of utilities — see §B.32 and the All-inclusive rate glossary entry.
VAT	UAE Value Added Tax. 5% standard rate applied to every line item charged to the guest on DSS — accommodation (Homes/Long-stay/Extended-stay), cleaning fees, Experiences, Services, and DSS commission. The only exception is the Tourism Dirham Fee (government not a taxable supply). DET-licensed Holiday Homes & SR are commercial/hospitality supplies — the Article 45 residential exemption does Not apply (v8.2 correction). DSS collects VAT at booking confirmation, remits quarterly to FTA. Corrected v8.2 from earlier "Homes Exempt" treatment.
weekly	Stay classification 7–27 nights. Weekly rate, weekly housekeeping default. v7.4 §V.

v8.1 additions

TERM	DEFINITION
ABTest	Unified experimentation framework (§B.46). Sticky assignment via consistent hash. Variants minimum 2, maximum 8.
Attribution model	Method for assigning credit to marketing touchpoints. MVP: last-click. Phase 3: first-touch, linear, time-decay variants.
BlogPost	Editorial content (§B.37). Human-authored, MDX-formatted. Distinct from programmatic SEO pages (§E.7).
Campaign	Marketing campaign coordinating multi-channel outreach (§B.44). May use O or N promo codes; tracked via UTM parameters.
HelpArticle	Knowledge base entry (§B.49). Audience-scoped: internal_ops · host · guest · public. Replaces ad-hoc runbook documentation.
Incident commander	The platform user assigned as decision-maker during an active SRE incident. Decides severity, drives resolution, owns postmortem.
IncidentReport	SRE incident record (§B.48). 4 severity levels (sev1-sev4). Tracks MTTD, MTTA, MTTR.
Integration partner	Third-party technology vendor whose API DSS integrates with. Channex, Stripe, Tabby, SendGrid, Twilio, AWS, PriceLabs. Distinct from Provid (§B.3, marketplace Experience/Service supplier) and from the decommissioned Vendor entity.
MDX	Markdown + React components. Format for BlogPost and HelpArticle body content. Supports embedded interactive components.
MTTD / MTTA / MTTR	Mean Time To Detect / Acknowledge / Resolve. SRE incident lifecycle metrics tracked on IncidentReport.
NotificationRule	Configuration for what events trigger notifications (§B.41). Replaces hardcoded notification dispatch logic.
Quiet hours	Recipient-level do-not-disturb window. Defined on User/Owner. Respected by non-critical NotificationRules.
SearchInsight	Aggregated finding derived from SearchQueryLog (§B.43). Surfaces supply gaps, zero-result patterns, conversion outliers.
SearchQueryLog	Persisted search query record (§B.42). 90 days hot, 2 years cold archive. Feeds Phase 2 OpenSearch ranking signals.
SEV1 / SEV2 / SEV3 / SEV4	Incident severity classification. SEV1 = full outage; SEV2 = major degraded; SEV3 = partial; SEV4 = minor.
SLO	Service Level Objective. Measurable target (e.g. p95 booking API < 200ms). Breaches trigger IncidentReport.
UTM	Urchin Tracking Module parameters (utm_source, utm_campaign, utm_medium, utm_content, utm_term). Marketing attribution tags appended to URLs.

Zero-result query	Search query returning 0 properties. Surfaced as SearchInsight to drive supply-gap acquisition decisions.
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v8.2 additions

TERM	DEFINITION
All-inclusive rate	The DSS pricing principle locked in v8.2: every rate displayed to a guest covers utilities, internet, district cooling, and all routine operating costs. Guests pay the rate. Hosts set rates that cover their operating exposure. DSS does not track, bill, mediate, or display utility consumption. See §B.32 for the full statement.
Host operating cost	Any cost the host incurs to run their property — utilities, internet, cleaning, maintenance, insurance, DET licensing fees. None of these are T concepts; hosts price them into the rate. Distinguished from <i>marketplace fees</i> (DSS commission, payment processing) which are DSS concepts.
DET classification	The Dubai DET-recognised accommodation class a property holds. One of four values DSS supports: <code>standard_holiday_home</code> · <code>deluxe_holiday_home</code> · <code>standard_sr</code> · <code>deluxe_luxury_sr</code> . Determined by the host's DET licence; stored as <code>Property.det_classification</code> drives TDF rate lookup.
SR (Serviced Residence)	DSS shorthand for the DET "Hotel Apartment / Serviced Residence" accommodation class. Renamed from "Hotel Apartment" in v8.2 for clarity. Two SR subclasses: Standard SR and Deluxe/Luxury SR. Different from Holiday Home — a separate DET licence category with different TDF rates.
TDFRateSchedule	Effective-dated reference table holding the 4×6 DET TDF matrix (§B.50). Admin-editable; two-person approval; full audit log. Replaces hardcoded TDF rates so DET schedule updates don't require a code release. Bookings store <code>schedule_version</code> for idempotent recalculation.
Property size (TDF input)	One of six values driving TDF lookup: <code>studio</code> , <code>br_1</code> , <code>br_2</code> , <code>br_3</code> , <code>br_4</code> , <code>br_5</code> . Properties with 6+ bedrooms use the <code>br_5</code> row pending DET clarification.

Total: ~94 terms (v8.0: 70 + v8.1: 18 + v8.2: 6). If a term in the SRS is unclear and not listed here, file an SRS amendment request per §O.5.

0.4 Engineering pickup order

Read this section first if you're new to the SRS. It tells you which sections to read in which order to start contributing. The full SRS is structured for completeness, not for onboarding — this section bridges the gap.

0.4.1 — Day 1: Orient yourself

- 1. §0.1 Abstract — understand DSS in 60 seconds
- 2. §0.2 Scope & status — what's in / what's out
- 3. §0.3 Glossary — bookmark this; you'll return to it
- 4. §A.1 Brand tokens — internalize colors, fonts, voice
- 5. §Z.3 ADRs locked — the 14 binding decisions; read the compendium

Time: ~2 hours. Outcome: you can discuss the platform coherently in a kickoff meeting.

0.4.2 — Week 1: Build the foundational mental model

Read these in order. They define how the entire booking lifecycle works:

- 1. §B.0 Property — the listing object
- 2. §B.13 User + KYC LO/L1/L2 model
- 3. §B.12 Owner — the host entity
- 4. §B.9 Booking — the core commerce entity
- 5. §O Booking lifecycle — 12-state state machine
- 6. §B.10 Payment + §B.11 Refund
- 7. §P Payment flow + §Q Refund engine
- 8. §B.16 DisputeCase + §B.17 PayoutBatch
- 9. §B.30 InsurancePolicy + §B.31 Claim + §U Claim lifecycle
- 10. §B.14 Review + §S Communications
- 11. §T Tax & legal — VAT + TDF + Pattern B

Time: 6 hours. Outcome: you understand everything from search, \$book, \$user, \$book, to \$review, \$refund

Time: ~6 hours. Outcome: you understand every step from search → book → pay → check-in → review → payout.

o.4.3 — Week 2: Subdomain you'll be working on

Pick one subdomain based on your assigned track. Read the relevant chapters in depth, including §M priorities and locked rules:

IF YOU'RE WORKING ON...	READ THESE SECTIONS
Public marketplace UI	§A Brand · §B.0–B.8 (property/area/POI/amenity/accessibility/city) · §D Search · §E Page architecture · §F Empty-state rules · §J SearchQuery · §K URL state
Booking + payment	§B.9 Booking · §B.10 Payment · §B.11 Refund · §O lifecycle · §P payment flow · §Q refund engine · §B.15 BookingModification · v6.4.1 §G Q.8 edge cases · ADR-006 (Stripe/Tabby)
Host cockpit	§B.12 Owner · §B.13 User KYC · §B.0 Property (host-side fields) · §B.23 RatePlan · §B.24 PricingRule · §S Communications · §T Tax (Pattern B authorisation)
Channel manager integration	§7 OTA integration · §8 Sync engine · ADR-001 · <code>DSS-CM-Integration-Spec.md</code> · §B.16 DisputeCase (channel-sourced conflicts)
Admin operations	<code>DSS-Admin-Dashboard-v3.html</code> · §B.16 DisputeCase · §B.17 PayoutBatch · §B.30/B.31 Insurance & Claims · §V Long-stay lifecycle · §4 RBAC
Long-stay flows Decommissioned v8.3	Long-stay scope removed v8.3. §V section deleted. §B.33–B.36 entities decommissioned. All bookings are short-stay.
Infrastructure / DevOps	§1–§3 Multi-tenancy · §9 Database · §11 DevOps · §13 SLOs · §14 Compliance · ADR-003, ADR-004, ADR-008, ADR-010, ADR-011, ADR-014
Data / analytics	§B-erd Entity relationships · §15 AI roadmap · §16 Reporting · ADR-013 AI sequencing

Time: ~8 hours. Outcome: you can implement your first feature without needing to read sections outside your subdomain.

o.4.4 — Reference, not read-through

These sections you reference when needed; don't try to read end-to-end:

- **§B-erd Entity relationships** — visual ERD; consult when designing joins
- **§Z.2 Entity backlog** — status of all 33 entities
- **§Z.4 Suggested next bumps** — pending SRS work
- **Companion documents:** `DSS-Pre-Development-Audit.md` · `DSS-Development-Plan.md` · `DSS-ADR-Compendium.md` · `DSS-CM-Integration-Spec.md` · `DSS-Insurance-Underwriter-RFP.md`

o.4.5 — The reading anti-pattern

Don't read the SRS top-to-bottom in order. You'll burn 20+ hours and lose track of the foundational concepts. Follow §0.4.1 → §0.4.2 → §0.4.3 instead. Roughly 16 hours of focused reading gets you productive; reference the rest as needed.

0.5 Amendment process

The SRS evolves. This section governs how. As of v8.0, the document is canonical and amendments follow a formal process.

0.5.1 — Change categories

CATEGORY	EXAMPLES	APPROVAL
Minor edit	Typo · clarifying sentence · broken link · cosmetic format	Any team member; peer review (1 reviewer)
Specification clarification	Better example · sharper acceptance criterion · expanded glossary entry	Engineering lead OR product lead
New entity / field / rule	Adding an entity · field on existing entity · new locked rule	Engineering lead + product lead
ADR amendment	Reversing or significantly modifying a locked ADR	New ADR superseding the old; founder + engineering lead sign-off
Brand or commercial change	Commission rate · brand tokens · DET licence relationship · entity decommissioning	Founder approval required
Phase reassignment	Moving an entity/feature from one phase to another	Engineering lead + product lead (with founder informed)

0.5.2 — Amendment lifecycle

1. **Propose** — open an issue in the SRS repository tagged `srs-amendment` with: section affected, change proposed, rationale, downstream impact
2. **Discuss** — async discussion for 48–72 hours; tagged stakeholders weigh in
3. **Decide** — approver(s) per §0.5.1 sign off
4. **Apply** — PR against the SRS HTML; reviewed by 1 peer + automated checks (tag balance, brand compliance)
5. **Version bump** — patch version increment (v8.0 → v8.0.1 for minor; v8.0 → v8.1 for new entity; v8.0 → v9.0 for major reframe)
6. **Communicate** — changelog entry in §0.6 Version history; engineering team notified

0.5.3 — Amendment frequency expectation

- **Phase 1 MVP build (weeks 0–12):** moderate amendment volume expected (~3–5 minor + 1–2 specification per week as engineering surfaces ambiguities)
- **Post-MVP launch:** low volume (~1–2 minor per week; specification changes rare)
- **Phase transitions:** material amendment activity expected (Phase 2 launch, Phase 3 SaaS multi-tenancy, Phase 4 internationalisation each likely a v8.2 / v8.3 bump)

0.5.4 — Locked items (require ADR amendment, not SRS amendment)

The following cannot be changed via SRS amendment; they require a new ADR superseding the relevant locked one:

- Backend framework choice (ADR-002)
- Database choice (ADR-003)
- Channel manager build-vs-buy (ADR-001)
- Multi-tenancy model (ADR-004)
- Region / data residency (ADR-014)
- Authentication provider (ADR-009)
- Commission tiers locked in §0.1 Abstract
- Brand tokens locked in §A.1
- Marketplace-vs-PMS positioning locked in v7.5 §1

Material changes to these items represent a redirection of the platform, not a refinement of the spec. They require founder approval and a formal ADI

0.6 Version history

VERSION	DATE	SCOPE
v6.0	baseline	Brand tokens, commercial rules, header/footer, Provider entity
v6.1	22 May 2026	Experience + Service + Provider entities · context-aware PillSearch · §F.1 honesty rules

v6.2	22 May 2026	Property + Area + POI + Amenity catalog (48) + City + \$J SearchQuery + \$K URL sync
v6.3	22 May 2026	SearchQuery extensions · map clustering · filter modal · view history · accessibility catalog (16)
v6.4	22 May 2026	Booking + Payment + Refund + Owner + User + Review + Communications + Tax — the transactional layer
v6.4.1	22 May 2026	BookingModification + DisputeCase + PayoutBatch + refund edge cases (chargebacks, partial captures, mid-installment Tabby, FX)
v7.0	22 May 2026	Enterprise architecture — PMS, Channel Hub, OTA integration, multi-tenancy, microservices, DevOps, compliance, AI, GCC expansion
v7.0.1	22 May 2026	Wishlist + PromoCode + Discount + RatePlan + PricingRule + EmailTemplate entities
v7.0.2	22 May 2026	HousekeepingTask + MaintenanceTicket + Vendor + CorporateAccount (operational entities — Vendor + Housekeeping + Maintenance subsequently decommissioned in v7.5)
v7.1	22 May 2026	Insurance + Claim entities (PROVISIONAL defaults — E1 strategic decision)
v7.2	22 May 2026	Native mobile app architecture (outline — Phase 5+)
v7.3	22 May 2026	Sovereign cloud variant (outline — trigger-dependent)
v7.4	22 May 2026	Long-stay & corporate workflows — UtilityCharge + SecurityDeposit + LeaseAgreement + RelocationPackage + InspectionReport entities \$V long-stay lifecycle
v7.5	22 May 2026	Marketplace scoping clarification — decommissioned HousekeepingTask + MaintenanceTicket + Vendor (operator concerns, not marketplace); InspectionReport renamed to DepositInspection (host+guest joint, marketplace-neutral); RelocationPackage reframed as marketplace Service bundle; TDF Pattern B (DSS collects + remits) locked
v8.0	22 May 2026	Canonical consolidation. Supersedes v6.1 through v7.5. Adds Glossary (§0.3), Engineering pickup order (§0.4), Amendment process (§0.5), Entity Relationship Diagram (§B-erd). Engineering reads this file as canonical; addenda preserved as <code>/srs/history/</code> reference only.
v8.1	22 May 2026	Platform-ops expansion. Adds 13 new entities across 7 subdomains: editorial CMS (BlogPost, EditorialAuthor, BlogCategory), notifications (Notification, NotificationRule), search insights (SearchQueryLog, SearchInsight), marketing (Campaign, CampaignAttribution), experimentation (ABTest, ABTestVariant), SRE (IncidentReport), knowledge base (HelpArticle). Entity count → 46. Renames "Vendor" → "Integration partner" / "Channex" / "Provider" depending on context — Vendor entity remained decommissioned per v7.5; the term itself now reserved for plain-English use.
v8.4	23 May 2026	Admin-gated channel distribution workflow. (A) <i>New entity B.52 ChannelDistribution</i> — per-property × per-channel record model the request → review → approve/reject → push → active lifecycle (10 states). Hosts <i>request</i> channel distribution; DSS admin <i>approves and selects</i> which channels mirror each property. Hosts cannot self-serve channel selection — DSS curates the marketplace AND its distribution footprint. Rationale: every Airbnb / Booking / Vrbo listing carries DSS brand reputation; channel-specific compliance (photo specs, policy formats, rate parity), economic suitability (commission stacking math), and anti-cannibalization are admin decisions. (B) <i>Admin decision criteria</i> formalized — channel-tier match, rate-parity math, anti-cannibalization, DET classification fit, compliance, strategic. Taxonomy-tagged decision reasons (11 tags) for analytics. (C) <i>SLA</i> — 7 standard, 48h signature, 24h re-submission after info required. (D) <i>Rate uplift</i> — admin sets <code>channel_rate_uplift_pct</code> per ChannelDistribution to absorb stacked commissions while preserving host net. (E) <i>Booking entity</i> gains optional <code>channel_distribution_id</code> FK for channel attribution and economics replay. (F) <i>Audit</i> — 10-action audit log per ChannelDistribution. Drift events from Channex (per ADR-001) surface on the same surface. Entity count: 43 → 44 (B.52 added). Composition: 28 commerce + 16 platform-ops = 44 active entities.
v8.3	23 May 2026	Two scope simplifications in one version. (A) <i>Long-stay scope removed entirely</i> . DubaiShortStay.com is a short-stay marketplace the name is literal. (1) \$V Long-stay lifecycle section deleted. (2) B.33 SecurityDeposit, B.34 ShortStayAgreement, B.35 RelocationPackage, B.36 DepositInspection — all four entities decommissioned (tombstones retained at §B.33–B.36). (3) Stay classifications <code>monthly</code> and <code>extended_stay</code> removed. All bookings are short-stay; <code>stay_classification</code> field dropped from Booking. (4) 28-night and 90-night thresholds deleted. Booking · <code>max_nights</code> default 27, admin-configurable per Property. (5) Damage protection consolidates onto DSS Coverage (§B.30/§B.31). (6) Phase 3+ long-stay product, if revived, would be a separate platform/SKU with its own SRS. (B) <i>Multi-tenancy removed entirely</i> . DSS is a single marketplace operated by Keys Please, not a SaaS platform sold to other operators. (1) §3 Multi-tenancy section rewritten as "Single-platform model" — hosts are accounts within the platform, not tenants. (2) ADR-004 (multi-tenancy: RLS + schema-per-tenant) superseded. Active ADR count: 14 → 13. (3) §4 RBA simplified: 13 platform roles + 3 host roles (was 8 tenant roles). (4) Tenant entity / <code>tenant_id</code> FK fields removed; data scopes to <code>host_id</code> instead. (5) Glossary "Tenant" entry decommissioned. (6) Phase 4 description updated — multi-tenant + GCC expansion

		GCC expansion alone. (/) If multi-tenancy ever revived (Spain white-label, sovereign-cloud), it would be a separate deployment, r folded back. Entity count: 47 → 43. Active ADRs: 14 → 13. Composition: 28 commerce + 15 platform-ops = 43 active entities.
v8.2	23 May 2026	(1) Locks the <i>all-inclusive rate principle</i> : rates include all utilities and operating costs. UtilityCharge entity (B.32) decommissioned <i>TDF schedule</i> replaced with authoritative DET 4-class × 6-size matrix, AED 10-100/night per property. "Hotel Apartment" renamed "SR" (Serviced Residence). (3) <i>New entity B.50 TDFRateSchedule</i> : effective-dated, admin-editable, two-person approval. (4) <i>VAT correction</i> : Homes from "Exempt" to 5% — DET Holiday Homes & SR are commercial supplies, not residential lettings. 5% applies uniformly across accommodation, cleaning, Experiences, Services, DSS commission. Only TDF is not vatable. New \$T.5 Money flow worked example. (5) <i>JARVIS renamed to MORCEL</i> throughout (AI suggestion engine). (6) <i>LeaseAgreement renamed to ShortStayAgreement</i> (SB.34) — reflects commercial hospitality nature; Ejari registration descoped (Ejari is for residential leases; C doesn't facilitate those). (7) <i>New entity B.51 CommissionSchedule</i> : admin-editable commission rates with min/max bands, two-person approval, per-Property/Experience/Service overrides. Replaces hardcoded 12/20/25%. Commercial model locked: hosts display GROSS rate (commission baked in), DSS deducts at payout. (8) Zero-result search handling formalized in SB.43.1 — 5-stage graceful fallback. Entity count: 46 → 47 (B.51 added). Composition: 32 commerce + 15 platform-ops = 47 active entities.

PART I

Foundations

Brand, design system, entity contracts, locked commercial rules. The immutable core that every subsequent decision builds on.

§A Brand & design system v6.0

A.1 Locked tokens

These tokens are immutable. Any UI artifact must use these exact values. Variations seen in the field (e.g., navy #082940 in admin prototypes) are violations and must be remediated.

TOKEN	VALUE	USAGE
Navy	#0D1F35	Primary, headings, navigation
Gold	#C9A84C	Accent, CTAs, dividers
Cream	#FAF7F2	Default background
Section bg	#EEF5FB	Alternating sections
Serif	Cormorant Garamond	Headlines, hero copy, prices
Body	DM Sans	All body text, UI
Monospace	DM Mono	References, codes, eyebrows

A.2 Header & navigation

Three product tabs (Homes / Experiences / Services), scroll-collapse behaviour, 2px gold accent line, PillSearch (context-aware per active tab), hamt

drawer for utility links. Locked in v6.0.

A.3 Locked commercial rules

RULE	VALUE	NOTES
Commission · Homes	12% LOCKED	All-inclusive (photography, listing, distribution, ops). Owner.commission_pct default. Per Design Review §9 C1 - final policy lock pending
Commission · Experiences	20% LOCKED	Provider pays commission on gross booking value
Commission · Services	25% LOCKED	Provider pays commission on gross booking value
Payout schedule	D+1 per check-in LOCKED	Net of commission, sent to verified bank account
Service response SLA	2 hours LOCKED	Auto-refund on breach. Services only.
Tourism Dirham	AED 10 → 100 per property per night LOCKED	DET 4-class × 6-size matrix (Standard/Deluxe Holiday Home, Standard/Deluxe-Luxury SR × Studio/1BR/2BR/3BR/4BR/5BR). Admin-editable TDFRateSchedule (§B.50). 30-night cap. See §T.2
VAT · Homes	5% LOCKED v8.2	DET-licensed Holiday Homes & SR are commercial/hospitality supplies — standard rated. Article 45 residentia exemption does NOT apply (corrected v8.2)
VAT · Experiences	5% LOCKED	Standard rated
VAT · Services	5% LOCKED	Standard rated (cleaning, concierge, transfers, etc.)
VAT · Cleaning fees	5% LOCKED	Charged on the booking, standard rated
VAT · DSS commission	5% LOCKED	DSS commission is a B2B service to the host; subject to standard VAT
VAT · TDF	Not vatable LOCKED	TDF is a government fee, not a taxable supply

§B Core entities

B.0 Property v6.2

The Home product. 60+ field schema with frozen pricing snapshot, immutable booking references, status lifecycle (draft → pending_review → live → paused → archived).

```
type Property = {
  // - Identity -
  id:                UUID;
  status:            "draft" | "pending_review" | "live" | "paused" | "archived";
  slug:              string;                                // SEO-friendly URL

  // - Content -
  title:              string;
  hero_photo:         URL;
  gallery:            URL[];                                // min 8 for live
  description:        string;
  property_type:      "apartment" | "villa" | "townhouse" | "penthouse";

  // - Capacity -
  bedrooms:           int;
  bathrooms:          float;
  max_guests:         int;
  beds:               { type, count }[];

  ..
}
```

```

// - Location -
area_id:          UUID;                // FK → Area §B.4
city_id:          UUID;                // FK → City §B.7
lat:             float;
lng:             float;
address_disclosed: boolean;           // false until booking confirmed

// - Pricing -
base_price_aed:   int;
cleaning_fee_aed?: int;
currency:         "AED";
is_instant_book:  boolean;            // default true

// - Catalog refs -
amenity_ids:      UUID[];             // from §B.6 (48 locked)
accessibility_ids: UUID[];            // from §B.8 (16 locked)

// - Regulatory -
det_permit_number: string;            // required for status=live
tdf_per_bedroom:  int;               // 7/10/15/20

// - Owner -
owner_id:         UUID;               // FK → Owner §B.12
commission_pct:   int;               // inherited from Owner

// - Cancellation -
cancellation_policy: "flexible" | "moderate" | "strict";

// - Curation -
active_badges:     [{type: "top", category: "home", expires_at}, {type: "dubai_favorite", category: "home"}, {type: "top_provider"}]
quality_score:     87 // 0-100 UQS;

// - Performance (read-only, computed) -
rating?:          float;              // null when review_count < 3 (honesty rule)
review_count:     int;
bookings_total:   int;

// - Lifecycle -
created_at:       ISO8601;
published_at?:    ISO8601;
};

```

B.1 Experience V6.1

Curated experiences across 6 categories. Provider-led but DSS-curated. Tier system (signature / editor's pick / standard) drives placement on listing pages.

FIELD	NOTES
<code>category</code>	<code>adventure</code> · <code>nautical</code> · <code>cultural</code> · <code>culinary</code> · <code>wellness</code> · <code>nightlife</code>
<code>provider_id</code>	FK → Provider §B.3
<code>duration_minutes</code>	Length of the experience
<code>pricing_unit</code>	<code>per_guest</code> · <code>per_group</code>
<code>meeting_point</code>	Fixed location with lat/lng
<code>permits_required</code>	Adventure + Nautical require DTCM; others Insurance minimum LOCKED
<code>is_instant_book</code>	Default <code>true</code>
<code>commission_pct</code>	Locked at 20%

B.2 Service V6.1

On-demand services with the 2-hour response SLA. Eight service types. Provider name and response-time badge appear on listing cards (unlike Experiences).

FIELD	NOTES
-------	-------

<code>service_type</code>	<code>housekeeping</code> · <code>private_chef</code> · <code>transfer</code> · <code>in_home_spa</code> · <code>grocery</code> · <code>childcare</code> · <code>personal_shopper</code> · <code>laundry</code>
<code>response_window_minutes</code>	Default 120 LOCKED
<code>auto_refund_if_breached</code>	Default true LOCKED
<code>escalation_path</code>	<code>ops_team</code> · <code>manual</code> · <code>auto_match_next_provider</code>
<code>pricing_unit</code>	<code>per_session</code> · <code>per_hour</code> · <code>per_guest</code>
<code>location_semantics</code>	<code>in_home</code> · <code>at_venue</code> · <code>delivery</code> · <code>anywhere</code>
<code>is_instant_book</code>	Default <code>false</code> (most services need confirm)
<code>commission_pct</code>	Locked at 25%

B.3 Provider **V6.1**

Single entity covers Experience and Service operators. KYC, payout, performance metrics. Distinct from Owner (§B.12) — Owner is for Homes.

B.4 Area **V6.2**

Dubai neighbourhoods. Geo polygon, hero copy, insider_data block (peak season, average nightly rate, busiest months). Drives the /dubai/[area] landi pages.

B.5 POI · Points of Interest **V6.2**

Landmarks, beaches, malls, dining. Used for distance-to-X queries and area-page sidebars.

B.6 Amenity catalog **V6.2** **48 LOCKED**

Read-only reference data. 48 amenities across 6 categories (Essentials, Living, Kitchen, Outdoor, Safety, Premium). Properties tag from this catalog or

B.7 City **V6.2**

Top-level geographic container. Currently Dubai only; Abu Dhabi, Riyadh, Jeddah, Doha, Valencia at Phase 4+.

B.8 Accessibility catalog **V6.3** **16 LOCKED**

Read-only reference data. 16 accessibility features across Mobility, Sensory, Communication, Cognitive categories. Drives the accessibility filter.

B.9 Booking **V6.4**

The transaction record. One booking per stay/experience/service. Polymorphic via `product_type`. Immutable `product_snapshot` and `pricing` aft creation.

```
type Booking = {
  // - Identity -
  id:                UUID;
  reference:         string;                // "DSS-2026-184729"
  status:            BookingStatus;        // see §C.4
  status_history:    StatusTransition[];

  // - Product (polymorphic, immutable) -
  product_type:      "Home" | "Experience" | "Service";
  product_id:        UUID;
  product_snapshot:  { title, photo, area_id?, cancellation_policy };

  // - Parties -
  guest_id:          UUID;                 // FK → User §B.13
  owner_id?:         UUID;                 // for Homes
  provider_id?:      UUID;                 // for Experiences/Services

  // - Dates -
  data_window:       { start: TS08601 and?: TS08601 duration minutes? 1.
```

```

date_window:      { start: 1000001, end: 1000002, duration_minutes: 1,
guests_adults:    int;
guests_children:  int;
guests_infants:   int;                                // not counted toward max_guests

// - Pricing snapshot (frozen at creation, never recalculated) -
pricing: {
  currency:        "AED";
  base_amount:     int;
  cleaning_fee?:   int;
  service_fee:     int;
  tdf?:            int;                                // Homes + SR only

  vat_aed:         int;                                // 5% standard, every line item except TDF (v8.2)
  discount?:       { code, type, amount };
  total:           int;
  owner_payout:    int;                                // pre-computed for Homes
  dss_take:        int;                                // commission + service_fee
};

// - Policy (snapshot at creation) -
cancellation_policy: "flexible" | "moderate" | "strict";
is_instant_book:     boolean;
rtb_expires_at?:     ISO8601;                        // 24h for RTB

// - Channel -
channel:             "direct" | "airbnb" | "booking_com" | "vrbo" | "expedia" | "agoda";
channel_reference?:  string;

// - Links -
payment_id?:         UUID;
refund_ids:          UUID[];
review_id?:          UUID;
thread_id?:          UUID;

// - Trust & regulatory -
dss_coverage_active: boolean;                        // E1 blocker - see §Z.1
det_permit_used?:    string;                          // required for Home + status=CONFIRMED
guest_id_verified:   boolean;                          // L1+ required for bookings > AED 5,000

// - Lifecycle -
created_at:          ISO8601;
confirmed_at?:       ISO8601;
check_in_at?:        ISO8601;
check_out_at?:       ISO8601;
completed_at?:       ISO8601;
cancelled_at?:       ISO8601;
};

```

B.10 Payment v6.4

One booking can have multiple payments (deposit + balance, or installments via Tabby). One payment can have multiple refunds. Hold/capture mode Card; immediate capture for Tabby + Bank.

```

type Payment = {
  id:                UUID;
  booking_id:        UUID;
  sequence:          int;                                // 1 = first, 2 = balance, etc.
  method:            "card" | "bank_transfer" | "tabby" | "tamara" | "wallet";
  provider:          "stripe" | "checkout_com" | "tabby" | "tamara" | "ziina" | "wallet_internal";
  amount:            int;                                // in AED minor units (fil)
  currency:          "AED" | "USD" | "EUR" | "GBP";
  amount_aed_equivalent: int;                            // for FX
  status:            PaymentStatus;                      // see §C.5

  // - Hold / capture model -
  authorised_amount?: int;
  authorisation_expires_at?: ISO8601;                    // typically 7 days for cards
  captured_amount?:    int;
};

```

```

// - 3DS / SCA -
three_ds_required:    boolean;
three_ds_status?:    "challenged" | "passed" | "failed" | "frictionless" | "exempt";

// - Tabby / Tamara installments -
installment_plan?:    InstallmentPlan;                // 4 installments typically

// - Bank transfer specifics -
bank_reference?:    string;
bank_confirmed_by?:    UUID;                        // admin user_id, manual reconciliation

// - Provider -
provider_transaction_id: string;
provider_metadata?:    JSON;

// - Retry -
attempt_number:    int;
parent_payment_id?:    UUID;                        // for retries

failure_code?:    string;
failure_message?:    string;
created_at:    ISO8601;
};

```

B.11 Refund v6.4

One payment can have multiple refunds. Multi-component breakdown (base / cleaning / TDF / VAT / service-fee / discount). Method-specific handling (Tabby cancels future installments).

🛡️ **Integrity rule:** `refund_amount + retained_amount` MUST equal the original payment's `captured_amount`. If a partial refund leaves residual liability, second refund record is created later — never modify the first.

B.12 Owner v6.4

Distinct from Provider. Owners do Homes. KYC documents (7 types), encrypted bank accounts, tax residency, withholding.

KYC DOCUMENT TYPE	REQUIRED FOR
<code>passport</code>	All hosts
<code>emirates_id</code>	UAE residents
<code>trade_license</code>	Company hosts
<code>title_deed</code>	Property ownership proof
<code>ejari</code>	Tenancy contract (if not owner)
<code>noc</code>	NOC from building owner (some buildings)
<code>poa</code>	Power of Attorney (non-resident)

🛡️ **Locked rule:** A property cannot transition to `status: live` unless its primary owner has `kyc_status: verified` AND a verified primary bank account.

B.13 User · Guest v6.4

3-tier KYC progression (LO / L1 / L2) with transaction limits. Tokenised payment methods (PAN never touches our infra). Corporate TRN support for VAT invoicing.

LEVEL	REQUIREMENTS	BOOKING LIMIT
LO	Signup only	Cannot book
L1	Email + phone verified	Up to AED 5,000

L2	Passport or Emirates ID verified	No limit
----	----------------------------------	----------

B.14 Review v6.4

Multi-axis scores (1–5 in .5 steps). 14-day submission window post-checkout. Verified-stay flag. Moderation pipeline (auto-flag → human review → approve/remove).

SCORE AXIS	APPLIES TO
overall	All (required, headline)
cleanliness	Homes
accuracy	Homes + Experiences
communication	All
location	Homes
check_in	Homes
value	All

B.15 BookingModification v6.4.1

Proposal-based modification flow (date change, guest count, upgrades). Counterparty must accept; 48h proposal window; auto-availability check at acceptance.

FIELD	NOTES
type	date_change · guest_count_change · upgrade · downgrade · duration_extend · duration_shorten
proposed_changes	Snapshot of new field values
price_delta	Computed: original_total, new_total, delta, requires_payment, requires_refund
status	PROPOSED · COUNTER_OFFERED · ACCEPTED · DECLINED · EXPIRED · CANCELLED · COMPLETED
expires_at	48 hours from proposal
availability_check_passed	Re-checked at acceptance time, not proposal time

 **Locked rules:** Modifications cannot be applied unilaterally. Flexible-policy guests get one free date change within cancellation window. Channel-sourced bookings need source-channel acceptance.

B.16 DisputeCase v6.4.1

Opened when `Booking.status` transitions to `DISPUTED`. Tracks reason, evidence, assignment, SLA, outcome, financial resolution.

Dispute reason codes

CODE	DESCRIPTION
property_misrepresentation	Listing didn't match reality
cleanliness_unacceptable	Arrival condition
safety_issue	Hazardous condition
amenity_unavailable	Listed amenity not working
host_no_show	No check-in possible
guest_property_damage	Host claim against guest

<code>guest_property_damage</code>	most claim against guest
<code>guest_house_rules_violation</code>	Smoking, parties, excess guests
<code>payment_dispute</code>	Chargeback initiated
<code>double_booking</code>	OTA sync failure
<code>force_majeure</code>	Weather, regulatory, war

SLA

- Standard priority: response 24h, resolution 7 business days
- Critical priority (chargeback / safety): response 4h, resolution per provider window
- Evidence window: 5 business days for both parties to submit
- Payout freeze: host payout for disputed booking frozen until closure

B.17 PayoutBatch v6.4.1

D+1 payout batches. Generated daily at 09:00 UAE for all bookings transitioning to `IN_PROGRESS` yesterday. Finance review required before bank submission.

Lifecycle

GENERATING → READY → REVIEWED (finance) → SUBMITTED (bank) →
PARTIALLY_SETTLED or SETTLED (per item) · FAILED (rare)

Locked rules: One currency per batch. Disputed bookings excluded. Withholding auto-registered for quarterly FTA remittance.

B.19-B.20 Wishlist + WishlistItem v7.0.1

Guest-curated saved-property lists. Multiple wishlists per user, with optional sharing.

FIELD	NOTES
<code>name</code>	User-chosen (default "My next trip")
<code>is_shared</code>	If true, accessible via shareable link
<code>collaborators</code>	Other users invited to edit
<code>items</code>	Polymorphic — Property, Experience, Service entries

B.21-B.22 PromoCode + Discount v7.0.1

Discount engine. Eligibility rules (first booking only / specific products / minimum stay / date windows / max redemptions). Discount frozen in booking pricing snapshot.

FIELD	NOTES
<code>code</code>	e.g. "WELCOME10"
<code>type</code>	<code>percent</code> · <code>fixed_amount</code> · <code>first_night_free</code>

<code>value</code>	10 (=10%) · 500 (=AED 500) · etc.
<code>eligibility_rules</code>	JSONB — <code>first_booking_only</code> , <code>product_ids</code> , <code>min_stay_nights</code> , <code>date_window</code> , etc.
<code>max_redemptions</code>	Per code; null = unlimited
<code>max_per_user</code>	Per user; default 1
<code>valid_from</code> / <code>valid_to</code>	Code activation window
<code>active</code>	Can be toggled off without deletion

B.23 RatePlan v7.0.1

The pricing structure attached to a Property. Base rate + seasonal overrides + day-of-week multipliers + length-of-stay discounts.

```
type RatePlan = {
  id:          UUID;
  property_id: UUID;
  base_nightly_aed: int;
  weekend_multiplier?: float;           // e.g., 1.15 for Fri/Sat
  seasonal_overrides: { start, end, nightly_aed }[];
  los_discounts:      { min_nights, discount_pct }[]; // length-of-stay
  last_minute_discount: { hours_before, discount_pct }[];
  occupancy_pricing?: { occupancy_pct_threshold, multiplier }[]; // dynamic
  channel_markups:    { channel, markup_pct }[]; // per-OTA adjustments
  is_active:          boolean;
};
```

B.24 PricingRule v7.0.1

Pluggable rules for dynamic pricing. AI suggestions (Phase 2–3) write to this entity.

B.25 EmailTemplate v7.0.1

Templated messages backing the 9 lifecycle triggers (§S.3). Per-locale variants. A/B testing on subject + CTA copy.

B.26–28 HousekeepingTask · MaintenanceTicket · Vendor **DECOMMISSIONED v7.5**

⚠ **Removed in v7.5 — marketplace scoping clarification.** DSS is a tech marketplace, not a property management company. Housekeeping is the host's responsibility; cleaning sold as a paid add-on uses the existing Service entity (§B.2) with 25% DSS commission. Maintenance complaints route through Communications (§S) + DisputeCase (§B.16, reason `amenity_unavailable` or `safety_issue`) + DSS Coverage (§B.30, trigger `host_no_show_alternative`). Vendors are the host's commercial relationships, outside DSS scope. See `DSS-SRS-Addendum-v7.5.md`.

B.29 CorporateAccount v7.0.2

B2B account for companies booking on behalf of employees. Phase 3+ entity. TRN required for VAT invoicing.

FIELD	NOTES
<code>company_trn</code>	UAE TRN — required, no exceptions
<code>authorised_user_ids</code>	Users allowed to book on this account
<code>requires_approval_above_aed</code>	Bookings above threshold need <code>approvers</code> sign-off
<code>billing_mode</code>	<code>per_booking</code> · <code>monthly-consolidated</code> · <code>quarterly-consolidated</code>
<code>contract_discount_pct</code>	0–15% negotiated

B.30 InsurancePolicy v7.1

 **PROVISIONAL.** Schema is locked. Specific values (cover amount, cap, underwriter, exclusions) are industry-standard defaults pending business confirmation with underwriter. Engineering can build; final values swap in via config without schema changes.

The active DSS Coverage policy. Encodes cover limits, deductibles, covered perils, exclusions, claim mechanics, payout method. Applies platform-wide not per-tenant (v8.3).

FIELD	PROVISIONAL DEFAULT	RANGE TO DISCUSS
<code>cover_per_incident_aed</code>	100,000	50,000 – 500,000
<code>annual_cap_per_property_aed</code>	500,000	250,000 – 1,000,000
<code>aggregate_cap_aed</code>	10,000,000	5M – 25M
<code>deductible_per_claim_aed</code>	500	0 – 2,000
<code>claim_window_hours</code>	72	48 – 168
<code>payout_sla_business_days</code>	14	7 – 30
<code>premium_per_property_monthly_aed</code>	50	25 – 150

Covered perils (defaults: all true)

Accidental damage · theft by guest · vandalism · fire/water damage · appliance breakage during stay · booking cancellation by DSS · host no-show alternative cost.

Exclusions (defaults: all true = excluded)

Pre-existing damage · intentional misuse by owner · normal wear and tear · acts of war/terrorism · valuables above AED 50,000 (provisional) · cash in currency · pets damage (unless pets-allowed property) · natural disasters (unless add-on).

B.31 Claim v7.1

A claim under an active InsurancePolicy. Auto-created from triggers, manually filed by owners, or escalated from DisputeCase outcomes.

Triggers (when claims are created)

TRIGGER	AUTO-CREATE?
<code>guest_caused_damage</code>	Manual (owner files)
<code>guest_theft</code>	Manual; police report required
<code>guest_vandalism</code>	Manual
<code>dispute_outcome_host_favor</code>	✓ Auto-DRAFT
<code>dss_cancellation</code>	✓ Auto-create + auto-submit
<code>host_no_show_alternative</code>	✓ Auto-create
<code>appliance_breakage_in_stay</code>	Manual

manual_ops_initiated

By ops on owner's behalf

B.32 UtilityCharge DECOMMISSIONED v8.2

- ⚠ **v8.2 decommission:** UtilityCharge entity removed. Same marketplace-positioning rationale as the v7.5 decommissioning of HousekeepingTask, MaintenanceTicket, and Vendor — these were property-operator concerns that don't belong in a marketplace platform.

The all-inclusive rate principle

All DSS rates are all-inclusive of utilities and operating costs. The guest pays the displayed rate. Utilities are a host operating cost — not a guest concern, not a DSS concept. The host is responsible for setting rates that cover their utility exposure.

This means:

- Guest never sees a utility line item · never receives a utility invoice · never authorises a utility charge
- Host accounts (DEWA, Etisalat, du, FEWA, SEWA, district cooling, internet) are *off-platform* — DSS has no relationship to them, does not see meter does not integrate with their APIs
- UtilityCharge entity removed from the schema. UC-#### identifier series retired. Any pre-v8.2 pilot data archives only — no active records
- The Booking entity stores a single all-inclusive rate. Property entity already exposes advertised_inclusions (§B.0); engineers display utilities as inclusive by default

Out of scope: Utility billing of any kind. Meter readings. Utility-provider APIs. Utility VAT collection. Utility dispute handling. Utility line items in invoices. Off-platform host guest utility settlements (DSS doesn't track these either — between host and guest privately if they exist, but expected zero under all-inclusive policy).

B.33 – B.36 Long-stay entities DECOMMISSIONED v8.3

B.33 SecurityDeposit · B.34 ShortStayAgreement · B.35 RelocationPackage · B.36 DepositInspection — all four long-stay entities decommissioned in v

- ⚠ **Out of scope — DSS is a short-stay marketplace.** DubaiShortStay.com — the name is literal. Long-stay rentals (monthly and extended_stay) sit in a different commercial frame: residential leases, Ejari, monthly invoicing, security-deposit cycles measured in months. Stretching the marketplace mox across both creates product confusion and operational complexity disproportionate to the marketplace value DSS provides. v8.3 removes long-stay scope entirely. All bookings are short-stay. Maximum stay duration is enforced as a Booking constraint (see §B.10 Booking · max_nights).

What this means concretely:

- Stay classifications monthly and extended_stay are removed. All bookings are short-stay. The stay_classification field is dropped from Booking (§B.10).
- SecurityDeposit (B.33) — removed. DSS Coverage (§B.30) is the only damage protection product. Hosts wanting a card-hold or escrow deposit must arrange off-platform; not DSS-mediated.
- ShortStayAgreement (B.34) — removed. The Booking record itself (§B.10) is the contract artefact. No separate signature flow.
- RelocationPackage (B.35) — removed. If individual relocation services (airport pickup, car rental, school admission) are wanted, they can be sold as separate Servi (§B.2). The "bundle" abstraction is dropped.
- DepositInspection (B.36) — removed. There is no SecurityDeposit to inspect against. Damage discovered post-checkout routes via Coverage claims (§B.31) + DisputeCase (§B.16) as for any short stay.
- The 28-night and 90-night thresholds are deleted. The only stay-length constraint is Booking · max_nights (default: 27 nights — admin-configurable per Proper
- §V Long-stay lifecycle (formerly an entire section) — removed.
- Out-of-scope confirmation: Ejari, long-stay agreement clauses, biweekly housekeeping for monthlies, mid-stay top-up payments, lease renewal reminders, RDC handling. None are DSS scope.

If long-stay is revived later (Phase 3+ as a separate product), it would be on a separate platform / SKU with its own SRS, not reintroduced into the short-stay marketplace SRS.

B.37 BlogPost v8.1

Editorial content surface — destination guides, market reports, host interviews, news. Distinct from programmatic SEO pages (§E.7) which are generated from structured data. BlogPost is human-authored, scheduled, indexable.

FIELD	TYPE	NOTES
<code>id</code>	uuid	PK
<code>slug</code>	varchar(120)	URL-safe · unique platform-wide (no tenant scope v8.3)
<code>title</code>	varchar(180)	SEO H1
<code>excerpt</code>	text	Meta description · social card
<code>body_mdx</code>	text	MDX content (Markdown + React components)
<code>hero_image_id</code>	uuid	FK · S3 asset
<code>author_id</code>	uuid	FK → §B.38 EditorialAuthor
<code>category_id</code>	uuid	FK → §B.39 BlogCategory
<code>tag_ids</code>	uuid[]	multi-tag
<code>status</code>	enum	<code>draft</code> · <code>scheduled</code> · <code>published</code> · <code>archived</code>
<code>scheduled_at</code>	timestamptz	nullable
<code>published_at</code>	timestamptz	nullable
<code>featured</code>	boolean	Hero placement flag
<code>linked_property_ids</code>	uuid[]	Editorial cross-link to Property entities
<code>linked_area_ids</code>	uuid[]	For destination guides
<code>linked_experience_ids</code>	uuid[]	For activity round-ups
<code>seo_meta</code>	jsonb	title override · description override · og_image · canonical
<code>reading_time_minutes</code>	int	auto-calculated
<code>view_count</code>	bigint	denormalized, eventual consistency

Out of scope at MVP: AI-assisted drafting, version control / revision history, multi-language translation (Phase 4 for Spain expansion), comment system. Workflow at MV single-author draft → publish; no editor approval queue. Approval queue is Phase 2.

B.38 EditorialAuthor **vs.1**

Author entity for BlogPost. Authors can be DSS staff (tied to User via `user_id`) or external contributors (freelance writers, hosted as Author records without User accounts).

FIELD	TYPE	NOTES
<code>id</code>	uuid	PK
<code>display_name</code>	varchar(80)	"By Sarah K."
<code>slug</code>	varchar(80)	URL: <code>/blog/author/sarah-k</code>
<code>user_id</code>	uuid	FK · nullable (external authors have no User)
<code>bio</code>	text	~150 words
<code>avatar_url</code>	varchar(500)	S3 path
<code>social_handles</code>	jsonb	twitter · linkedin · instagram
<code>is_active</code>	boolean	

B.39 BlogCategory v8.1

Categorization for BlogPost. Each post has exactly one primary category, plus arbitrary tags. Categories appear in URLs: `/blog/category/dubai-guid`

FIELD	TYPE	NOTES
<code>id</code>	uuid	PK
<code>name</code>	varchar(60)	"Dubai Guides"
<code>slug</code>	varchar(60)	"dubai-guides"
<code>description</code>	text	SEO description
<code>parent_id</code>	uuid	FK self · nullable · 1-level hierarchy max
<code>post_count</code>	int	denormalized

MVP categories (locked seed): Dubai Guides · Neighborhood Spotlights · Market Reports · Host Stories · News · Travel Tips · Long-Stay Insights · Corporate Travel.

B.40 Notification v8.1

Outbound message to a specific recipient (User or Owner). Triggered by NotificationRule (§B.41) on state events. Includes in-app, email, SMS, WhatsApp push (mobile Phase 3). One Notification = one delivery attempt per channel.

FIELD	TYPE	NOTES
<code>id</code>	uuid	PK
<code>recipient_id</code>	uuid	FK · User OR Owner
<code>recipient_type</code>	enum	<code>user</code> · <code>owner</code> · <code>platform_user</code>
<code>rule_id</code>	uuid	FK → §B.41 (nullable for ad-hoc)
<code>channel</code>	enum	<code>in_app</code> · <code>email</code> · <code>sms</code> · <code>whatsapp</code> · <code>push</code>
<code>template_id</code>	uuid	FK · EmailTemplate (§B.25) or other channel template
<code>subject</code>	varchar(200)	rendered from template
<code>body</code>	text	rendered (channel-appropriate)
<code>payload</code>	jsonb	structured data for in-app rendering
<code>trigger_event</code>	varchar(80)	e.g. <code>booking.confirmed</code>
<code>related_entity_type</code>	varchar(40)	
<code>related_entity_id</code>	uuid	
<code>status</code>	enum	<code>queued</code> · <code>sending</code> · <code>delivered</code> · <code>failed</code> · <code>bounced</code> · <code>read</code>
<code>provider</code>	varchar(40)	SendGrid · Twilio · Apple/Firebase
<code>provider_message_id</code>	varchar(200)	
<code>queued_at</code> · <code>sent_at</code> · <code>delivered_at</code> · <code>read_at</code>	timestampz	nullable
<code>error_detail</code>	jsonb	nullable · for failed

B.41 NotificationRule v8.1

Configuration for what events trigger which notifications. Replaces hardcoded notification logic in services. Recipient preferences (do-not-disturb hc channel preferences) are stored on User (§B.13) and Owner (§B.12) — this entity defines the default rule behaviour.

FIELD	TYPE	NOTES
<code>id</code>	uuid	PK

<code>name</code>	varchar(80)	"Booking confirmation"
<code>trigger_event</code>	varchar(80)	e.g. <code>booking.confirmed</code> · <code>dispute.opened</code>
<code>conditions</code>	jsonb	e.g. <code>{"booking.total_aed": ">=50000"}</code>
<code>recipient_role</code>	enum	<code>guest</code> · <code>host</code> · <code>both</code> · <code>platform_ops</code>
<code>channels</code>	varchar[]	<code>{email, sms}</code>
<code>template_ids</code>	jsonb	per-channel template references
<code>respect_quiet_hours</code>	boolean	defer to recipient's do-not-disturb
<code>respect_unsubscribe</code>	boolean	marketing rules: yes · transactional: no
<code>is_active</code>	boolean	
<code>priority</code>	enum	<code>critical</code> · <code>high</code> · <code>normal</code> · <code>low</code>

MVP rules locked: 9 lifecycle triggers per §S.3 are seeded as NotificationRule rows. Editing rules is restricted to `PLATFORM_ADMIN` role to prevent accidental message suppression.

B.42 SearchQueryLog v8.1

Persisted record of every search query for analytics, zero-result detection, ranking signal training. Not to be confused with the `SearchQuery` runtime defined in §J — that's transient; this entity is stored.

FIELD	TYPE	NOTES
<code>id</code>	uuid	PK
<code>user_id</code>	uuid	FK · nullable for anon
<code>session_id</code>	uuid	anon session anchor
<code>query_text</code>	text	raw input
<code>parsed_intent</code>	jsonb	area · dates · guests · filters extracted
<code>filters_applied</code>	jsonb	after user refinement
<code>result_count</code>	int	0 = zero-result
<code>top_results</code>	uuid[]	top 5 property IDs shown
<code>clicked_result_id</code>	uuid	nullable
<code>clicked_position</code>	int	nullable · 1-indexed
<code>converted_booking_id</code>	uuid	nullable · attribution
<code>timestamp</code>	timestampz	indexed

Retention: 90 days hot · 2 years cold archive for ML training. PII (user_id) is hashed for queries older than 30 days. Phase 2 feeds OpenSearch ranking signals (ADR-007 ADR-013).

B.43 SearchInsight v8.1

Analytics table for SearchInsight v8.1. Columns: `id` (PK), `user_id` (FK), `session_id` (FK), `query_text` (text), `parsed_intent` (jsonb), `filters_applied` (jsonb), `result_count` (int), `top_results` (uuid[]), `clicked_result_id` (uuid), `clicked_position` (int), `converted_booking_id` (uuid), `timestamp` (timestampz).

Aggregated insight derived from SearchQueryLog. Surfaced in the Search Insights admin module. Examples: "1,24 / searches for beach front 4BR last 30 results returned" (supply gap); "Top search 'Marina 2BR' converts at 14% — above average."

FIELD	TYPE	NOTES
id	uuid	PK
insight_type	enum	top_query · zero_result · supply_gap · conversion_outlier
query_signature	varchar(200)	normalized canonical query
volume_30d	int	
conversion_rate	decimal(5,4)	
recommended_action	text	human-readable recommendation
estimated_value_aed	decimal(12,2)	if gap closed
computed_at	timestamptz	nightly batch
status	enum	open · acknowledged · actioned · dismissed

B.43.1 Zero-result search handling vs. 2

When a guest's search returns zero matching properties, DSS does **not** show an empty page. The runtime returns a graceful response in five stages, captured in SearchQueryLog with a `fallback_stage` field for analytics:

- Stage 1 — Loosen filters automatically.** The runtime drops the most restrictive filter first (typically exact bedroom count or area sub-zone) and re-runs the search. If results return, the response is labelled *"Closest matches — we relaxed [filter]"* so the guest knows. `fallback_stage = filter_relaxed`.
- Stage 2 — Adjacent areas.** If still zero, expand the geo radius (e.g., Marina → JBR + Marina + JLT). *"Showing nearby areas."* `fallback_stage = geo_expanded`.
- Stage 3 — Date flexibility.** If still zero, suggest ± 3 days on the check-in date or ± 1 night on length. *"These dates are popular — try ± 3 days."* `fallback_stage = dates_flexible`.
- Stage 4 — Editorial fallback.** If still zero, surface (a) up to 6 hand-curated featured properties from the same DET classification, and (b) any related BlogPost (§B.37) with `linked_area_ids` matching the search. *"No exact matches — explore these instead."* `fallback_stage = editorial`.
- Stage 5 — Capture intent.** If still zero AND the user is signed in (or willing to provide email), offer *"Notify me when properties matching this become available"*. This creates a SavedSearchAlert (lightweight record on Wishlist §B.18 with `type = alert`). `fallback_stage = alert_offered`.

What the runtime never does: show an empty results page with no next action; auto-redirect to homepage; show generic "Try again" without specifics; fabricate properties; show properties that don't match the search to inflate inventory perception (this would harm trust and conversion).

B.43.2 Insight generation from zero-result queries

Every zero-result query — regardless of which fallback stage resolved it — writes to SearchQueryLog (§B.42) with `result_count = 0` and the parsed search intent. A nightly batch job aggregates by `query_signature`:

- If a normalized query signature accumulates ≥ 50 zero-result instances in 30 days, a SearchInsight is created with `insight_type = zero_result`.
- If the search intent decomposes to a (area, bedroom_count, property_class) tuple that has *fewer than 5 active listings*, the insight is promoted to `insight_type = supply_gap` with `estimated_value_aed = (volume_30d × estimated_avg_booking_value × estimated_conversion_rate)`.
- Supply gaps appear in the Search Insights admin module sorted by estimated value, driving inventory acquisition decisions ("we need more 4BR v. 3BR on the Palm").

B.43.3 MORCEL (AI) involvement

MORCEL (the DSS AI suggestion engine, renamed from JARVIS in v8.2 — see ADR-013) does not directly serve search results. The search ranking is deterministic (rule-based scoring, Phase 2 adds OpenSearch). MORCEL is involved in:

- Query parsing** — extracting structured intent from natural language ("beach apartment for 4 people next weekend under 1500" → `{ area_feature: 'beach', guests: 4, dates: relative_weekend, max_rate_aed: 1500 }`)

- **Spell correction** — "Marrina 2br" → "Marina 2BR"
- **Synonym expansion** — "beach front" matches "beachfront", "sea view", "ocean view" (curated synonym dictionary)
- **Editorial fallback selection** — choosing the 6 most contextually-appropriate featured properties when Stage 4 triggers (semantic similarity to the parsed intent)
- **Insight recommendation text** — generating the human-readable `recommended_action` on SearchInsight records ("Acquire 4BR Palm villa invento 1,247 searches in last 30d, est. AED 142K/mo lost GMV")

MORCEL acceptance metrics are tracked per surface (query parsing, spell, synonyms, editorial picks) and surfaced in the Admin · MORCEL Intelligence module. Acceptance rates target 50-80% per surface — too high suggests over-conservative thresholds (missed opportunities), too low suggests pc model fit.

B.44 Campaign **vs.1**

Marketing campaign coordinating multi-channel outreach (email, social, paid, content). Tracks budget, channels, attribution. Distinct from PromoCode (\$B.21) — a campaign may use 0 or N promo codes.

FIELD	TYPE	NOTES
<code>id</code>	uuid	PK
<code>name</code>	varchar(120)	"Summer Europe 2026"
<code>type</code>	enum	<code>acquisition</code> · <code>retention</code> · <code>seasonal</code> · <code>corporate</code> · <code>long_stay</code>
<code>start_date</code> · <code>end_date</code>	date	
<code>budget_aed</code>	decimal(12,2)	
<code>spent_aed</code>	decimal(12,2)	running total
<code>channels</code>	varchar[]	<code>{email, instagram, google_ads, content}</code>
<code>target_segment</code>	jsonb	guest filter criteria
<code>promo_code_ids</code>	uuid[]	associated promo codes
<code>utm_source</code>	varchar(60)	tracking
<code>utm_campaign</code>	varchar(120)	
<code>status</code>	enum	<code>draft</code> · <code>active</code> · <code>paused</code> · <code>completed</code> · <code>archived</code>
<code>owner_id</code>	uuid	FK · platform user (Revenue Lead)

B.45 CampaignAttribution **vs.1**

Records which campaign drove a Booking. One Booking can have multiple attribution records (multi-touch). Default attribution model at MVP: last-cli Phase 3 introduces first-touch, linear, time-decay alternatives.

FIELD	TYPE	NOTES
<code>id</code>	uuid	PK
<code>booking_id</code>	uuid	FK → §B.9
<code>campaign_id</code>	uuid	FK → §B.44
<code>touchpoint_type</code>	enum	<code>first</code> · <code>last</code> · <code>assist</code>
<code>touchpoint_at</code>	timestamptz	when guest interacted
<code>channel</code>	varchar(40)	email · instagram · google · direct
<code>utm_params</code>	jsonb	raw utm_*
<code>attribution_weight</code>	decimal(5,4)	0.00–1.00 per model
<code>attributed_revenue_aed</code>	decimal(12,2)	

B.46 ABTest **vb.1**

Unified A/B test framework. Replaces test-specific ad-hoc logic. Tests can target: pricing rules, email subject lines, search ranking variants, promo co values, listing layouts. Assignment is sticky per user via consistent hash.

FIELD	TYPE	NOTES
<code>id</code>	uuid	PK
<code>name</code>	varchar(120)	"Review request subject line"
<code>target_surface</code>	enum	<code>email_subject</code> · <code>pricing</code> · <code>search_rank</code> · <code>promo_value</code> · <code>page_layout</code>
<code>hypothesis</code>	text	Recorded before launch
<code>start_date</code> · <code>end_date</code>	timestamptz	
<code>traffic_allocation</code>	decimal(3,2)	% of eligible users · 0.10 = 10%
<code>audience_filter</code>	jsonb	e.g. guest segment
<code>primary_metric</code>	varchar(60)	e.g. <code>click_rate</code> · <code>conversion</code> · <code>revenue_per_user</code>
<code>guardrail_metrics</code>	varchar[]	must not regress
<code>minimum_detectable_effect</code>	decimal(5,4)	statistical power param
<code>status</code>	enum	<code>draft</code> · <code>running</code> · <code>analysis</code> · <code>complete</code> · <code>shipped</code> · <code>rolled_back</code>

winner_variant_id	uuid	FK → §B.47 · nullable
-------------------	------	-----------------------

B.47 ABTestVariant v8.1

Variants of an ABTest. Minimum 2 (control + treatment); supports up to 8 variants (multi-arm). One must be flagged as control.

FIELD	TYPE	NOTES
id	uuid	PK
test_id	uuid	FK → §B.46
name	varchar(60)	"Variant B"
is_control	boolean	exactly 1 per test
config	jsonb	variant payload (subject line · price modifier · etc.)
traffic_share	decimal(3,2)	within-test allocation
impressions · conversions	bigint	denormalized counters
primary_metric_value	decimal(12,4)	running rate

B.48 IncidentReport v8.1

SRE incident record. Triggered by SLO breach, on-call paged, or post-hoc declaration. Drives the System Health admin module timeline. Distinct from DisputeCase (§B.16, customer-facing) — this is infrastructure-facing.

FIELD	TYPE	NOTES
id	uuid	PK
ref	varchar(20)	"INC-0042"

title	varchar(200)	short summary
severity	enum	sev1 (full outage) · sev2 (major degraded) · sev3 (partial) · sev4 (minor)
status	enum	open · investigating · identified · monitoring · resolved · postmortem
triggered_by	enum	slo_breach · on_call_page · customer_report · manual
affected_components	varchar[]	booking · payments · channel-sync · etc.
started_at	timestampz	
detected_at	timestampz	MTTD calculation
acknowledged_at	timestampz	MTTA
resolved_at	timestampz	MTTR
commander_id	uuid	FK · incident commander (platform user)
impact_summary	text	users affected · bookings lost
root_cause	text	after postmortem
postmortem_url	varchar(500)	Confluence / Notion link
action_items	jsonb	follow-up tasks

B.49 HelpArticle v8.1

Knowledge base entry — internal ops runbooks AND public guest/host help center articles. Visibility controlled by audience field.

FIELD	TYPE	NOTES
id	uuid	PK
slug	varchar(120)	URL
title	varchar(180)	
body_mdx	text	MDX content
category	varchar(60)	"Booking" · "Payments" · "Long-stay" · "Disputes" · etc.
tags	varchar[]	
audience	enum	internal_ops · host · guest · public
linked_runbook	varchar(120)	e.g. "CM drift > AED 100/day" links here
author_id	uuid	FK · platform user
last_reviewed_at	timestampz	auto-expires after 180 days for runbooks
helpful_yes · helpful_no	int	public articles only
view_count	bigint	denormalized
status	enum	draft · published · stale · archived

Platform-ops entity range: B.37 through B.49 (13 entities) were grouped as v8.1 platform-ops additions. v8.2 adds B.50 TDFRateSchedule and B.51 CommissionSchedule (admin-editable reference tables with two-person approval).

B.50 TDFRateSchedule v8.2

Effective-dated reference table for Tourism Dirham fee rates. Replaces hardcoded TDF rates in services. Each record represents one DET schedule version — the 4x6 matrix (4 property classes x 6 size tiers). When DET publishes a revised schedule, Platform Admin creates a new record with the new `effective_from` date; bookings query the schedule effective at `booking.created_at`.

Why this is an entity, not a config flag: DET rate changes are rare but real. Historical bookings must retain the rate they accrued at, for audit and consistency. A single mutable config value would silently rewrite history. An effective-dated table preserves the audit trail and lets Finance Ops run “\would TDF have been under the old schedule” comparisons.

FIELD	TYPE	NOTES
<code>id</code>	uuid	PK
<code>schedule_version</code>	varchar(40)	"DET-2026-Q2" · "DET-2014-baseline" · admin-assigned
<code>effective_from</code>	date	Inclusive · indexed
<code>effective_to</code>	date	Exclusive · nullable for current/open-ended schedule
<code>rate_matrix</code>	jsonb	4x6 matrix as nested object · see structure below
<code>night_cap</code>	int	Default 30 · DET-mandated cap on consecutive nights
<code>refund_policy</code>	enum	<code>no_checkin_only</code> (current) · others reserved
<code>vat_treatment</code>	enum	<code>not_vatable</code> (current) · TDF is a government fee, not a taxable supply
<code>source_document_url</code>	varchar(500)	Link to DET circular / regulation PDF that defines this schedule
<code>created_by</code>	uuid	FK · platform user (PLATFORM_ADMIN or FINANCE_ADMIN)
<code>approved_by</code>	uuid	FK · platform user · second signature for compliance
<code>created_at</code>	timestamptz	
<code>approved_at</code>	timestamptz	Required for <code>status = active</code>
<code>status</code>	enum	<code>draft</code> · <code>pending_approval</code> · <code>active</code> · <code>superseded</code> · <code>cancelled</code>
<code>notes</code>	text	Admin notes · e.g. "DET schedule update per Resolution 12 of 2026"
<code>open_items</code>	jsonb	Edge cases noted in admin: e.g. "6+ BR pending DET clarification"

B.50.1 rate_matrix structure

```
{
  "standard_holiday_home": {
    "studio": 10.00, "br_1": 10.00, "br_2": 20.00,
    "br_3": 30.00, "br_4": 40.00, "br_5": 50.00
  },
  "deluxe_holiday_home": {
    "studio": 15.00, "br_1": 15.00, "br_2": 30.00,
    "br_3": 45.00, "br_4": 60.00, "br_5": 75.00
  },
  "standard_sr": {
    "studio": 10.00, "br_1": 10.00, "br_2": 20.00,
    "br_3": 30.00, "br_4": 40.00, "br_5": 50.00
  },
  "deluxe_luxury_sr": {
    "studio": 20.00, "br_1": 20.00, "br_2": 40.00,
    "br_3": 60.00, "br_4": 80.00, "br_5": 100.00
  }
}
```

```

    "DI_3": 60.00, "DI_4": 60.00, "DI_5": 100.00
  }
}

```

All amounts in AED, per property per night. Validation rule: every (class, size) cell must have a value > 0 before `status` can transition to `active`. Properties with 6+ bedrooms use the `br_5` value pending DET clarification (logged in `open_items`).

B.50.2 Audit and history

Every TDFRateSchedule change is logged to the global audit log (§B.13 User audit entries are user-scoped; tax schedule changes log to a platform-scoped audit channel). Required fields on every audit entry: `actor_user_id` · `action` (`create` · `edit` · `approve` · `activate` · `supersede`) · `before` · `after` · `justification_text`. Activation requires two-person approval (creator ≠ approver).

B.50.3 Lookup contract

Services compute TDF for a booking via this lookup contract:

```

function calculateTDF(booking: Booking): TDFCalc {
  const schedule = await tdfRateScheduleRepo.findEffectiveOn(booking.created_at)
  const cls = booking.property.det_classification // enum
  const size = bedroomToSize(booking.property.bedroom_count) // size_studio ... size_br_5
  const ratePerNight = schedule.rate_matrix[cls][size]
  const cappedNights = Math.min(booking.total_nights, schedule.night_cap)
  return {
    schedule_version: schedule.schedule_version,
    rate_per_night: ratePerNight,
    nights_charged: cappedNights,
    nights_capped: booking.total_nights - cappedNights,
    total_aed: ratePerNight * cappedNights,
    vatable: false
  }
}

```

Idempotency: The computed `schedule_version` is stored on the Booking at confirmation so re-calculation always reproduces the original value, even if a new TDFRateSchedule has been activated.

B.51 CommissionSchedule v8.2

Effective-dated reference table for DSS commission rates across the three product lines (Homes, Experiences, Services). Replaces the previously-hardcoded 12/20/25% rates. Admin-editable with two-person approval, full audit log, and per-property/per-experience/per-service override capability. Same design pattern as TDFRateSchedule (§B.50).

Commercial model (locked v8.2): Hosts and providers display the **gross rate** on their listings — the price the guest will pay. The host has factored DSS commission into that price (e.g., for a desired AED 700 net at 12% commission, the host lists AED 795.45 ≈ AED 800 gross). At booking confirmation, DSS deducts commission from the gross rate before paying the host. Commission is not a separate guest-facing line item; it's absorbed in the displayed price. This is the standard B2C marketplace pricing model (Booking.com style, also Airbnb Plus).

B.51.1 Schedule entity

FIELD	TYPE	NOTES
<code>id</code>	uuid	PK
<code>schedule_version</code>	varchar(40)	"DSS-2026-launch" · "DSS-2026-Q3-adjustment" · admin-assigned
<code>effective_from</code>	date	Inclusive · indexed
<code>effective_to</code>	date	Exclusive · nullable for current
<code>homes_default_pct</code>	decimal(5,4)	e.g. 0.1200 (12%) · launch default
<code>homes_min_pct</code>	decimal(5,4)	Floor for property-level overrides · e.g. 0.0500
<code>homes_max_pct</code>	decimal(5,4)	Ceiling for overrides · e.g. 0.2500
<code>experiences_default_pct</code>	decimal(5,4)	e.g. 0.2000 (20%) · launch default
<code>experiences_min_pct</code>	decimal(5,4)	
<code>experiences_max_pct</code>	decimal(5,4)	
<code>services_default_pct</code>	decimal(5,4)	e.g. 0.2500 (25%) · launch default
<code>services_min_pct</code>	decimal(5,4)	
<code>services_max_pct</code>	decimal(5,4)	
<code>vat_on_commission</code>	boolean	Default true · 5% VAT applies to commission per v8.2 §T.1
<code>created_by</code>	uuid	FK · platform user (PLATFORM_ADMIN or FINANCE_ADMIN)
<code>approved_by</code>	uuid	FK · platform user · must differ from <code>created_by</code>
<code>created_at</code> · <code>approved_at</code>	timestampz	
<code>status</code>	enum	<code>draft</code> · <code>pending_approval</code> · <code>active</code> · <code>superseded</code> · <code>cancelled</code>
<code>notes</code>	text	"Launch defaults" · "Q3 promotional rate cut for Experiences"

B.51.2 Per-entity overrides

Each Property (§B.0), Experience (§B.1), and Service (§B.2) carries an optional commission override:

```
Property.commission_override_pct    decimal(5,4) // nullable
Property.commission_override_reason text          // admin justification, required if override set
Property.commission_override_set_by uuid           // FK platform user
Property.commission_override_set_at timestampz

Experience.commission_override_pct  decimal(5,4) // nullable
Experience.commission_override_reason text
... (same shape)

Service.commission_override_pct    decimal(5,4) // nullable
... (same shape)
```

Override rules: Must fall within the `[min_pct, max_pct]` band on the active `CommissionSchedule`. Setting an override requires `commission_override_reason` (free text indexed for audit search). All overrides logged to platform audit channel. Removing an override (set to NULL) requires similar justification.

B.51.3 Lookup contract

```
function getCommissionPct(item: Property|Experience|Service, asOf: Date): CommissionLookup {
  const schedule = await commissionScheduleRepo.findEffectiveOn(asOf)
  const default_pct = pickDefault(schedule, item.kind) // homes / experiences / services
  const effective_pct = item.commission_override_pct ?? default_pct
  return {
    schedule_version: schedule.schedule_version,
    // ...
  }
```

```
default_pct,
override_pct: item.commission_override_pct,
effective_pct,
vat_applies: schedule.vat_on_commission
}
}
```

Idempotency: Booking stores `commission_schedule_version` AND `commission_pct_applied` at confirmation, so recalculations always reproduce the original commission even after schedule changes or override edits.

B.51.4 Launch defaults & bands

PRODUCT LINE	DEFAULT %	MIN %	MAX %	NOTES
Homes (HH and SR)	12%	5%	25%	Same default applies to all 4 DET classes; per-property overrides where strategically justified
Experiences	20%	10%	30%	Per-provider and per-experience overrides supported
Services	25%	15%	35%	Higher take-rate reflects DSS T&S overhead on real-time services

Min/max bands prevent fat-finger errors on the override surface. Bands themselves are admin-editable on the schedule — changing them requires the same two-person approval as activating a new schedule.

B.51.5 Audit and two-person approval

Every CommissionSchedule and per-entity override change is logged: `actor_user_id` · `action` (`create` · `edit` · `approve` · `activate` · `superse` `override_set` · `override_remove`) · `before` · `after` · `justification_text` . Schedule activation requires two distinct admins (creator ≠ approve

Per-entity overrides can be set by a single admin within the band, but appear on a daily reconciliation report sent to FINANCE_ADMIN for review.

B.52 ChannelDistribution **vs. 4**

Per-property × per-channel distribution record. Models the explicit, admin-gated workflow for mirroring a DSS property to an external OTA channel (Airbnb, Booking.com, Vrbo, Agoda, Expedia, Trip.com, Google Vacation Rentals, etc.).

⚠ **v8.4 workflow lock:** Hosts *request* channel distribution; DSS admin *approves and selects* which channels each property mirrors to. Hosts cannot self-serve channel selection. This is intentional — DSS curates the marketplace AND its distribution footprint. Every Airbnb / Booking / Vrbo listing carries DSS brand reputation. Channel-specific compliance (photo specs, policy formats, rate parity), economic suitability (commission stacking math), and anti-cannibalization are admin decisions.

B.52.1 Entity schema

FIELD	TYPE	NOTES
<code>id</code>	uuid	PK
<code>property_id</code>	uuid	FK → Property (SB.O). One row per property × channel pair. Indexed.
<code>channel</code>	enum	<code>airbnb</code> · <code>booking_com</code> · <code>vrbo</code> · <code>agoda</code> · <code>expedia</code> · <code>trip_com</code> · <code>google_vr</code> · <code>marriott_homes_villas</code> · <code>plum_guide</code> (extensible — new enum values per ADR amendment)
<code>status</code>	enum	10 states · see B.52.2 lifecycle. Indexed for queue filtering.
<code>requested_by</code>	uuid	FK → User · the host-side user who initiated the request
<code>requested_at</code>	timestamptz	Submission timestamp · drives SLA clock
<code>requested_notes</code>	text	Host's justification — why mirror to this channel (optional but encouraged)
<code>requested_target_net_aed</code>	decimal(10,2)	Host's target net per night (optional) — admin uses this to assess if rate uplift on the channel can preserve host economics after stacked commissions
<code>reviewed_by</code>	uuid	FK → User · platform admin who decided (PLATFORM_ADMIN or OPS_MANAGER role required)
<code>reviewed_at</code>	timestamptz	Decision timestamp · used for SLA compliance reporting
<code>decision_reason</code>	text	Required when status = <code>rejected</code> ; optional when <code>approved</code> · taxonomy-tagged for analytics (see B.52.4)
... (10 more fields)		

<code>channel_rate_uplift_pct</code>	decimal(5,4)	Admin-set rate multiplier for this channel (e.g., 0.18 for 18% uplift to absorb Airbnb's 15% commission + maintain host net). Applied on top of host's DSS gross rate.
<code>cancellation_policy_override</code>	varchar(40)	Channel-specific policy if different from property default (some channels require their own scheme)
<code>external_listing_id</code>	varchar(80)	Channel-side ID after successful first push (nullable until <code>active</code>)
<code>external_url</code>	text	Public listing URL on the channel (nullable until <code>active</code>)
<code>last_synced_at</code>	timestamptz	Last successful Channex bidirectional sync
<code>drift_flagged</code>	boolean	True when Channex detected price/availability/content drift between DSS and the channel (per ADR-001)
<code>drift_count_30d</code>	int	Rolling 30-day count — high drift triggers admin review
<code>paused_at</code>	timestamptz	If currently paused, when
<code>paused_by</code>	uuid	FK → User (admin OR host) — host can pause own listing on a channel; admin can pause any
<code>paused_reason</code>	text	Free text · taxonomy-tagged
<code>archived_at</code>	timestamptz	Soft delete · rows preserved for audit even after host withdrawal or channel removal
<code>created_at</code>	timestamptz	

B.52.2 Lifecycle — 10 states

STATUS	MEANING	SET BY	NEXT STATES
<code>not_requested</code>	Implicit default · no row exists for this property × channel	—	<code>requested</code>
<code>requested</code>	Host submitted a request for this channel	Host	<code>under_review</code> · <code>withdrawn</code>
<code>under_review</code>	Admin opened the request · SLA clock active	Admin	<code>approved</code> · <code>rejected</code> · <code>info_required</code>
<code>info_required</code>	Admin sent question back to host (e.g., need landscape hero photo)	Admin	<code>under_review</code> (host responded) · <code>withdrawn</code>
<code>approved</code>	Admin approved · awaiting Channex push to channel	Admin	<code>active</code> (push succeeded) · <code>push_fail</code>
<code>push_failed</code>	Channex push failed (channel rejected payload, schema mismatch, etc.) · ops triage	Channex	<code>approved</code> (retry) · <code>rejected</code> (give up)
<code>active</code>	Live on the channel · Channex syncing	Channex	<code>paused</code> · <code>archived</code>
<code>paused</code>	Temporarily inactive · host or admin paused	Host OR Admin	<code>active</code> (resume) · <code>archived</code>
<code>rejected</code>	Admin denied · reason recorded · host can submit new request	Admin	<code>archived</code>
<code>withdrawn</code>	Host withdrew request before admin decision	Host	<code>archived</code>

Note: `archived` is not a status — it's an `archived_at` timestamp. Archived rows remain queryable for audit; they don't appear in active queues.

B.52.3 Admin decision criteria

When admin opens a `requested` row, they review:

CHECK	WHAT ADMIN VERIFIES
Property DSS-approved	Property.status = <code>active</code> on DSS. Cannot mirror something not live on DSS itself.
Channel-tier match	Is this property's badge status suitable for the channel? Premium channels (Plum Guide) require Top badge or Dubai Favorite status.
Channel-ready content	Photos match channel spec (e.g., Airbnb wants landscape hero $\geq 1024\text{px}$; Booking wants room-by-room shots). Property description meets character minimums per channel.
Rate parity workable	After stacked commissions (DSS + channel), can the host's <code>requested_target_net_aed</code> still be met with a sensible <code>channel_rate_uplift_pct</code> ? If not, reject with explanation.
Anti-cannibalization	If property already at 80%+ occupancy on direct DSS traffic, mirroring may split availability without revenue lift. Admin may delay or reject.
DET classification fit	Some channels treat <code>SR</code> differently from <code>holiday_home</code> . Admin confirms the channel's category mapping is correct.
Cancellation / deposit policy fit	Channel-required policies don't conflict with DSS defaults. Override field used if needed.
Compliance	DET permit valid · KYC L1+ · no open DisputeCase against host with this property · no recent T&S flag
Strategic	Does this distribution serve broader DSS goals (Airbnb superhost ladder, Booking Genius enrollment, partnership leverage)?

B.52.4 Decision reason taxonomy

Decision reasons are taxonomy-tagged for analytics (free text supplemented with a tag from this list):

TAG	WHEN
<code>approved_standard</code>	Routine approval, no special notes
<code>approved_signature_uplift</code>	Signature property pushed for strategic visibility
<code>approved_with_uplift_adjustment</code>	Admin set non-default <code>channel_rate_uplift_pct</code>
<code>rejected_photos_below_spec</code>	Photos don't meet channel requirements
<code>rejected_rate_economics</code>	Stacked commissions don't preserve viable host net
<code>rejected_cannibalization</code>	Property too strong on direct; mirroring not value-additive
<code>rejected_compliance</code>	DET / KYC / outstanding T&S issue

<code>rejected_badge_mismatch</code>	Property does not meet channel's badge requirements (Top or Dubai Favorite)
<code>rejected_strategic</code>	DSS strategic reason (e.g., partnership negotiation hold)
<code>info_required_photos</code>	Need additional / different photos
<code>info_required_policy</code>	Need cancellation policy clarification
<code>info_required_other</code>	Other clarification needed

B.52.5 SLA & queueing

TIER	TARGET	ESCALATION
Standard property channel request	72h	Notification to OPS_MANAGER at 60h
Signature property channel request	48h	Notification to OPS_MANAGER at 36h
Re-submission after <code>info_required</code>	24h	Faster — most review already done
Admin-initiated pause	Immediate	Channex sync triggered same minute
Drift event resolution	24h	Auto-pause if <code>drift_count_30d</code> ≥ 5

B.52.6 Rate uplift math & persistence

```
// Compute channel-side display rate
function computeChannelRate(property, channel_distribution) {
  const dss_gross = property.gross_rate_aed // host's listed price on DSS
  const uplift = channel_distribution.channel_rate_uplift_pct // admin-set
  const channel_gross = dss_gross * (1 + uplift)
  return Math.round(channel_gross) // pushed to channel via Channex
}

// Example: Marina 2BR · DSS gross AED 850 · approved for Airbnb with 18% uplift
//   Airbnb display: AED 1,003
//   Guest pays AED 1,003 on Airbnb
//   Airbnb takes ~15% commission = AED 150 → AED 853 to DSS
//   DSS books AED 850 to property + AED 3 channel margin (or 0, by parameter)
//   DSS commission applied to host: 12% × 850 = AED 102
//   Host nets AED 748 (same as a direct DSS booking at 850 gross)

// Bookings store channel_distribution_id at confirmation for full audit trail.
```

Idempotency: Booking entity gains `channel_distribution_id` (nullable — null = direct DSS booking). Once recorded at confirmation, the booking always replays with the channel-side economics that applied at the time, even if the ChannelDistribution row is later paused or has a new `uplift_pct`.

B.52.7 Audit log

Every status transition is logged: `actor_user_id` · `action` (`request` · `open` · `approve` · `reject` · `info_request` · `resubmit` · `pause` · `resume` · `withdraw` · `archive`) · `before_status` · `after_status` · `reason`. Surfaced on the Channel Manager admin module and on the property-specific log (§B.O audit trail).

B.52.8 Foreign key on Booking

Booking (§B.IO) gains an optional `channel_distribution_id` FK. When a booking originates from an external channel (not direct DSS), this FK points to the ChannelDistribution row that brought the guest in. Used for:

- Channel-attribution reporting (which channel converts best per property class)
- Channel-specific economics replay (Booking permanently records the `uplift_pct` and channel commission at the time)
- Coverage and dispute routing — which channel's policy applies to which booking

Platform-ops entity range update: B.50 TDFRateSchedule + B.51 CommissionSchedule + B.52 ChannelDistribution. Total platform-ops = 16. **Active entity count:** 44 (28 commerce + 16 platform-ops).

§2.1.1 Entity Relationship Diagram

Visual map of the 33 entities grouped by domain, with the most important foreign-key relationships drawn. Not every FK is shown — entities like **Booking** have ~15 outbound FKs which would obscure the diagram. Domain groupings reflect how engineering should think about the schema: **Product** (what's listed), **Consumer** (who books), **Commerce** (the transaction), **Settlement** (the money), **Communications** (how parties interact), **Insurance** (DSS Coverage), **Long-stay** (mediation entities for ≥28 night stays).



B-crd.1 — Cardinality notes

- Property : Booking** — 1:N. One property hosts many bookings over time. No overlap permitted (calendar enforces).
- User : Booking** — 1:N as guest. A user can have many bookings; one user per booking as primary guest.
- Owner : Property** — 1:N. One owner manages many properties; a property belongs to one owner.
- Booking : Payment** — 1:N. A booking can have multiple Payment records (initial hold + final capture + add-on captures + modification top-ups).
- Payment : Refund** — 1:N. A payment can be partially refunded multiple times.
- Booking : DisputeCase** — 1:N. Multiple disputes possible per booking (rare, but cleanliness + payment chargeback could co-exist).
- Booking : Review** — 1:1. One review per booking, post-checkout, within 14-day window.
- SecurityDeposit : DepositInspection** — 1:N. Move-in + move-out = 2 typical inspections per deposit.
- RelocationPackage : Service** — M:N. A package bundles multiple Services; a Service can appear in multiple packages.

B-erd.2 — Domain ownership (engineering tracks)

DOMAIN	ENGINEERING OWNERSHIP	PHASE
Product	Marketplace / Public web team	Phase 1
Consumer	Identity / Account team	Phase 1
Commerce	Booking + Payments team (highest-stakes)	Phase 1
Settlement	Finance / Ops team	Phase 1
Insurance	Trust & Safety team (pending E1)	Phase 1 (provisional) → Phase 1.1 (final values)
Long-stay mediation	Marketplace + Finance teams jointly	Phase 1 (initial) → Phase 2 (depth)

§C Enumerations

C.4 Booking status V6.4 12 STATES

STATUS	MEANING
PENDING	Created but unconfirmed (RTB only)
PAYMENT_PENDING	Host/provider accepted, awaiting guest payment
CONFIRMED	Paid (or hold placed), locked in
IN_PROGRESS	Guest checked in / experience started / service began
COMPLETED	Successful, awaiting review window
CANCELLED_BY_GUEST	Guest cancelled; refund per §C.6 schedule
CANCELLED_BY_HOST	Host/provider cancelled; 100% refund + DSS Coverage
CANCELLED_BY_DSS	DSS-initiated (SLA breach, fraud, DET issue); 100% refund
DECLINED	RTB declined by host/provider
EXPIRED	RTB timed out (24h)
DISPUTED	Either party opened a dispute
RESOLVED	Dispute closed

C.6 Cancellation policy schedules V6.4 LOCKED

Frozen at booking creation. Even if the host changes the policy later, the booking honours the original.

POLICY	WINDOW	REFUND
Flexible	≥ 24h before	100% of base, less service fee
	< 24h before	50% of first night, rest refunded

	Day-of / no-show	No refund
Moderate	≥ 5 days before	100% of base, less service fee
	1–5 days before	50% of base, less service fee
	< 24h	No refund
	No-show	No refund
Strict	≥ 14 days before	50% of base, less service fee
	< 14 days	No refund
	No-show	No refund

Overrides (apply to all policies)

- **Host cancels:** 100% refund of total + DSS Coverage applies
- **DSS cancels:** 100% refund of total
- **SLA breach** (Services): Auto-100% refund per [§B.2](#)
- **Force majeure:** 100% refund of base, less service fee — case-by-case

Fee treatment

FEE	GUEST CANCELS	HOST/DSS CANCELS
Service fee	Retained	Refunded
Cleaning fee	100% if ≥ 24h before	Refunded
TDF	100% if no check-in only	100% if no check-in only
VAT (Services)	Proportional	100%

§D Context-aware search · PillSearch v6.1

The search bar switches second-field semantics per active tab:

TAB	FIELD 1	FIELD 2	FIELD 3
Homes	Where	When	Guests
Experiences	Where	When	Guests
Services	Where	Type of service	Guests

§E Page architecture v6.1

- **E.1 Property Detail** — 10 sections, has map view
- **E.2 Experience Detail** — 10 sections, has map view
- **E.3 Service Detail** — 9 sections, no map (services come to the guest)

§F Empty-state rules v6.1

- F.1 Honesty rule (locked):** aggregate ratings are displayed publicly only when the reviewed entity has ≥ 3 approved reviews. Below 3, display `bookings_total` ("New on DSS · 12 bookings") instead. Never invent or estimate a rating.

F.2 Search empty states: Specific messaging for zero-result searches with suggested broadening actions (drop a filter, try nearby area, alternative d

PART II

Discovery

Search query model, filtering, map clustering, URL state. How guests find properties.

§J SearchQuery & filters V6.2 V6.3

The query model used by listing pages, map views, and the partner API.

```
type SearchQuery = {
  city?:      UUID;
  area?:      UUID;
  bbox?:      [west, south, east, north];    // for map drag

  date_window?: { start, end };
  guests?:      { adults, children, infants, pets };

  // - Filters -
  property_type?: "apartment" | "villa" | "townhouse" | "penthouse";
  bedrooms_min?:  int;
  bathrooms_min?: float;
  price_min?:     int;
  price_max?:     int;
  amenity_ids?:   UUID[];                    // from §B.6
  --
```

```

accessibility_ids?: UUID[]; // from $B.8
cancellation?: "flexible" | "moderate" | "strict";
host_languages?: string[];
house_rules?: { pets, parties, smoking };
is_instant_book?: boolean;
required_badges?: ["top", "dubai_favorite"] // null = any badge/no badge;

// - Sort -
sort?: "relevance" | "price_low" | "price_high" | "rating" | "newest";

// - Pagination -
page?: int;
per_page?: int; // default 24
};

```

J.4 Map clustering

Properties cluster at low zoom, expand into individual price-pins at high zoom. Click cluster → zoom in or list view.

J.5 Search-as-map-moves

Toggle on the map view. When enabled, panning/zooming the map updates the result list. Default off (per Airbnb).

J.6 Filter modal

Single modal with all filters grouped: Place type, Price, Bedrooms, Amenities (collapsible by category), Accessibility, Cancellation, House rules, Booking type.

§K URL state v6.2

Every filter change updates the URL via History API (no reload). Shareable URLs. Back-button returns to prior search state. Example:

```

/dubai/downtown?check_in=2026-06-18&check_out=2026-06-23
&guests=2&bedrooms_min=2&price_max=2500
&amenities=pool,gym&cancellation=flexible

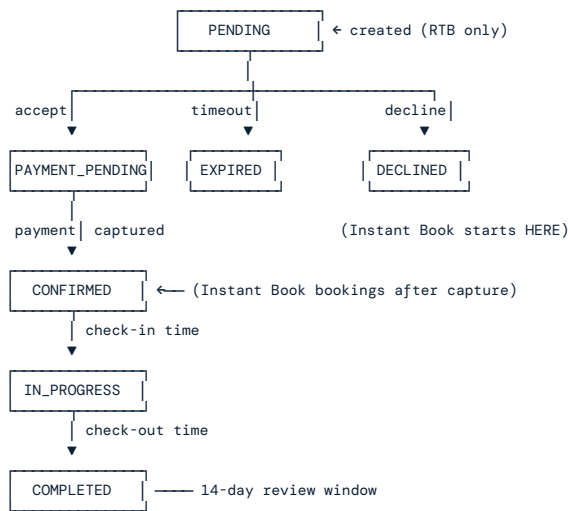
```

The *transactional* lifecycle

From the Reserve button to the D+1 payout. State machine, payment flows, refund engine, reviews, tax handling.

§0 Booking lifecycle v6.4

O.1 State machine



At ANY pre-COMPLETED state:

- CANCELLED_BY_GUEST (follows §C.6 schedule)
- CANCELLED_BY_HOST (always 100% refund)
- CANCELLED_BY_DSS (always 100% refund)

→ DISPUTED (opens ticket)

Terminal: COMPLETED · EXPIRED · DECLINED · CANCELLED_* · RESOLVED

O.2 Transition side-effects

TRANSITION	SIDE EFFECTS
* → CONFIRMED	Send guest confirmation; notify host; block calendar; allocate DET permit reference; create CommunicationThread
* → IN_PROGRESS	Send pre-arrival pack (Homes); release smart-lock code; start DSS Coverage; trigger TDF accrual
* → COMPLETED	Schedule payout (D+1); review-request email (T+24h, T+7d); release security deposit hold
* → CANCELLED_BY_GUEST	Compute refund per §C.6; create Refund record; unblock calendar; notify host
* → CANCELLED_BY_HOST	100% refund; apology email + alternatives; penalise host cancellation rate; credit goodwill voucher to guest
* → DISPUTED	Freeze payout; open ops ticket; assign case manager; collect evidence

O.3 RTB timeout

Host has 24h from `created_at` to accept/decline. Reminders at T+12h and T+22h. Auto-expire at T+24h. Instant Book skips PENDING entirely.

SP Payment flow v6.4

P.1 Card payment (default)

Default flow: **authorise at booking** → **capture at check-in** (Homes/Experiences) or **authorise + capture at booking** (Services where work begins immediately).

1. Guest hits Reserve
2. Frontend creates PaymentIntent with Stripe / Checkout.com
3. Guest enters card; 3DS challenge (almost always required in UAE per CBUAE)
4. On success: Payment status `AUTHORISED`, Booking status `CONFIRMED`
5. At check-in time: capture the hold → status `CAPTURED`
6. Capture failures retry T+30min, T+2h, then mark FAILED + 24h guest grace period

For stays > 7 days from booking, re-authorise at T-5d (cards expire holds at 7 days).

P.2 Tabby / Tamara (BNPL)

4-installment plan. Tabby pays DSS upfront; assumes credit risk on guest. Refund routed through Tabby.

Restriction: Tabby — UAE/KSA/KWT only. Tamara — UAE/KSA/EGY. Both disabled for bookings > AED 30,000.

P.3 Bank transfer

Displays IBAN + booking reference. Payment status stays `AWAITING_CONFIRMATION` until ops manually reconciles (typically 1–2 business days). Bank

transfer disabled by default for bookings < AED 1,500.

P.4 3DS / SCA

3DS required for almost all UAE-issued cards (CBUAE). Frictionless flow when issuer signals low-risk. Exemptions only for repeat-customer transaction AED 100.

§Q Refund engine v6.4

Q.1 Calculation

```
function compute_refund(booking, trigger, requested_at):
    if trigger in ['host_cancel', 'dss_cancel', 'sla_breach']:
        return full_refund(booking)           # 100% of EVERYTHING

    if trigger == 'guest_cancel':
        schedule = booking.cancellation_policy
        hours_until_start = (booking.date_window.start - requested_at).hours
        return apply_schedule(booking, schedule, hours_until_start)

    if trigger == 'goodwill':
        return manual_refund(booking, admin_specified_amount)

    if trigger == 'fraud':
        return full_refund(booking) + block_guest_account()

    if trigger == 'chargeback':
        return mirror_chargeback(booking)
```

Q.3 Timing

METHOD	EXPECTED ARRIVAL
Card (original)	T+5 to T+10 business days
Tabby	Future installments cancelled immediately; refund of paid portion T+5d
Bank transfer	T+1 to T+3 business days (manual)
Wallet	Immediate

§R Reviews & ratings v6.4

Window: from `booking.completed_at` to `completed_at + 14 days`. One review per booking; no edits after submission.

R.3 Moderation pipeline

```
SUBMITTED
↓
automated checks (profanity · PII · length · language)
↓
clean & verified stay → PENDING queue → human review → APPROVED → PUBLIC
flagged              → REVIEW queue (priority) → REMOVED · or APPROVED w/ edit-request
```

Auto-flags

- Contains phone numbers, email addresses, URLs
- Body < 30 characters
- Detected language ≠ guest's profile language

- Profanity (per-locale word list)
- Score variance (5★ overall with < 30 chars body, or 1★ with no specifics)

§§ Communications v6.4

S.I Channels

Channel	Used for	Default opt-in
Email (transactional)	Confirmations, reminders, payment links	Mandatory
Email (marketing)	Newsletter, launches	Opt-in
SMS	OTP, urgent payment failures, check-in	On (can opt out)
WhatsApp	Pre-arrival, in-stay support, reviews	Opt-in
In-app	All transactional	On

S.3 Templated triggers 9 TEMPLATES

TRIGGER	TEMPLATE	CHANNEL (S)
Booking CONFIRMED	booking_confirmation	Email + WhatsApp + in-app
T-7 days before check-in	pre_arrival_info	Email + WhatsApp
T-1 day	check_in_reminder	Email + SMS
T+2h after check-in	arrival_followup	WhatsApp + in-app
T-24h before check-out	check_out_reminder	Email + in-app
T+24h after check-out	review_request_1	Email
T+7d after check-out	review_request_2	Email (if no review)
Cancellation	cancellation_confirmation	Email + in-app
Refund issued	refund_notification	Email

§T Tax & legal V6.4

T.I VAT (UAE 5%) CORRECTED V8.2

v8.2 correction: Earlier versions of this section listed Homes as VAT-exempt under Article 45 (residential lettings). That was wrong on the law. DET-licensed Holiday Homes and Serviced Residences are **commercial / hospitality supplies**, not residential lettings. They attract standard 5% VAT regardless of stay length. The Article 45 residential exemption applies only to standard Ejari long-term residential leases — which DSS does not facilitate. All product lines on DSS now uniformly attract 5% VAT.

Locked policy: Every line item the guest is charged for on DSS is subject to 5% standard-rated VAT, except the Tourism Dirham Fee (a government fee, not a taxable supply).

LINE ITEM	VAT RATE	NOTES
Homes (accommodation)	5%	Standard rated. Commercial supply via DET Holiday Home or SR licence — not residential. Applies to short-stay, monthly (28-1 nights), and extended_stay (≥90 nights) classifications equally.
Cleaning fees	5%	Standard rated. Charged as a line on the booking or as a separate Service — same treatment either way.

Experiences (all)	5%	Standard rated. Includes excursions, tours, activities, in-property catering, anything sold as an Experience (§B.1).
Services (all)	5%	Standard rated. Includes housekeeping, private chef, transfer, in-home spa, grocery, childcare, personal shopper, laundry, concierge — anything sold as a Service (§B.2).
DSS commission	5%	DSS commission to the host is a B2B service — subject to 5% VAT. Invoiced to the host (or netted from payout). See §T.5 for the commission/VAT flow.
Tourism Dirham Fee	Not vatable	Government fee, not a taxable supply (§T.2). Displayed as a separate line on the invoice with no VAT calculation.

T.1.1 Display convention

Per UAE Consumer Protection regulation, prices displayed to guests are **VAT-inclusive**. The marketplace listing card shows the all-inclusive nightly rate (operating costs + 5% VAT). At checkout, the invoice breaks out the components for transparency: net rate, VAT, TDF, any add-on services with their own VAT lines.

T.1.2 Registration & remittance

Keys Please Holiday Homes Rental Dubai LLC is VAT-registered with the FTA at MVP. DSS collects VAT at booking confirmation, holds it in a segregated tax liability account, and remits to FTA quarterly per the standard tax period assigned to the entity. When additional licensed hosts onboard (Phase 2+), the marketplace operates under the deemed-supplier model: DSS continues to be the VAT collection point of record for transactions through the platform, simplifying compliance for individual hosts who may not be VAT-registered (taxable supplies under AED 375,000/year).

T.1.3 Invoice format

Every guest receives a tax-compliant invoice (FTA Article 65 requirements) within 14 days of booking confirmation. Required fields: TRN of supplier (Keys Please), sequential invoice number, date, description of supply, taxable amount per line, VAT amount per line, total. Corporate guests with a `company_type` receive a fully tax-credit-eligible invoice; non-corporate guests receive a simplified tax invoice. See §T.3.

T.2 Tourism Dirham LOCKED (DET MATRIX · V8.2)

v8.2 schedule correction: Earlier SRS versions stored TDF as a 4-row per-bedroom heuristic (Studio/1BR/2BR/3BR/4BR+ → AED 7/10/15/20). That schedule was inaccurate. The authoritative DET schedule is a **4-class × 6-size matrix**, with rates from AED 10 (Studio · Standard Holiday Home / Standard SR) to AED 100 per night (5BR · Deluxe/Luxury SR). The matrix is now stored as an admin-editable reference table (§B.50 TDFRateSchedule) rather than hardcoded in services, so DET schedule updates can be applied without a code release.

The DET Tourism Dirham fee accrues per property per night based on two dimensions: **property class** and **property size**.

T.2.1 Property class taxonomy

DSS supports four DET-recognised accommodation classes for marketplace listings:

- Standard Holiday Home** — DET Holiday Homes licence, standard quality grade. Keys Please's anchor inventory falls under this class.
- Deluxe Holiday Home** — DET Holiday Homes licence, deluxe quality grade (premium fit-out, higher-tier amenities).
- Standard SR (Serviced Residence)** — DET Hotel Apartment / Serviced Residence licence, standard quality grade. Renamed from "Hotel Apartment" in v8.2 for clarity.
- Deluxe / Luxury SR (Serviced Residence)** — DET Hotel Apartment / Serviced Residence licence, deluxe or luxury quality grade.

Each host's DET licence determines which class their property falls into. The `Property.det_classification` field stores this (enum: `standard_holiday_home`, `deluxe_holiday_home`, `standard_sr`, `deluxe_luxury_sr`).

T.2.2 Rate matrix (effective from launch)

PROPERTY SIZE	STANDARD HOLIDAY HOME	DELUXE HOLIDAY HOME	STANDARD SR	DELUXE / LUXURY SR
Studio	AED 10	AED 15	AED 10	AED 20
1 BR	AED 10	AED 15	AED 10	AED 20
2 BR	AED 20	AED 30	AED 20	AED 40
3 BR	AED 30	AED 45	AED 30	AED 60
4 BR	AED 40	AED 60	AED 40	AED 80

5 BR	AED 50	AED 75	AED 50	AED 100
------	--------	--------	--------	---------

All rates are **per property per night** (not per bedroom — that's a different convention used by some hotels). Bedroom count is one input to the matrix lookup, not a multiplier. For prop with 6+ bedrooms, use the 5-BR row pending DET clarification — captured as an `open_item` on the schedule record.

T.2.3 Operating rules

- **30-night cap.** TDF accrues for a maximum of 30 consecutive nights per stay. Nights 31 onwards carry zero TDF. For long-stay (28–89 nights) and extended_stay (≥90 nights) classifications, the TDF total caps out regardless of stay length.
- **Per booking, not per guest.** The fee is per room/unit per night. Solo guests and family bookings pay the same TDF for the same property and date
- **Collected by DSS, remitted to DET.** Pattern B per v7.5 §7: DSS collects on the host's behalf at booking confirmation, remits to DET monthly via the operator account (Keys Please licence #727937 at MVP; per-host operator accounts as additional licensed hosts onboard).
- **Refund rule.** 100% refunded only if the guest never checked in. Partial stays (early check-out) generate no TDF refund — TDF for consumed nights retained per DET Article 7 (consumption-based).
- **VAT treatment.** TDF is a government fee, not a service. Not subject to 5% VAT. Displayed separately from the rate on the guest invoice.

T.2.4 Schedule administration

Rates are stored in **TDFRateSchedule (\$B.50)** as an effective-dated reference table — not hardcoded in services. When DET publishes a revised sch Platform Admin creates a new TDFRateSchedule record with the new `effective_from` date and the updated matrix. Bookings query the schedule effective at their `booking.created_at` date, so historical bookings retain their original TDF computation. The admin UI for this lives at **Admin · Finance TDF rates**.

T.3 Corporate invoicing

Guests with `is_corporate: true` and valid `company_trn` receive a tax-compliant VAT invoice. Sequential invoice numbering per FTA requirements generated within 24h of confirmation.

T.4 Non-resident owner withholding

For Owners with `tax_residency != "AE"`: DSS withholds per bilateral tax treaty schedule; remitted to UAE FTA quarterly. Owner receives withholdin certificate annually.

T.5 Money flow & VAT layering **v8.2**

The most-asked engineering question on this section: *what does the guest pay, and how is it split?*

T.5.0 The commercial model (locked v8.2)

Hosts and providers display the **gross rate** — the price the guest pays. DSS commission is *baked into* the gross rate; it is not a separate guest-facing The host has factored DSS commission and their own operating costs (utilities, cleaning, maintenance, margin) into the gross rate before listing.

Example: a host wants to net AED 700 per night after DSS commission of 12%. They list at AED 795.45 gross (≈ AED 800 rounded). The guest sees AE 800. At booking, DSS deducts AED 96 commission (12% of 800) + AED 4.80 commission VAT, and pays the host AED 699.20 net. The host's target net achieved within rounding.

Worked example for a 3-night Marina 2BR Deluxe Holiday Home at host-listed AED 800/night gross, plus a Service (cleaning) at AED 200 gross, sold t non-corporate guest. Commission rates per the active CommissionSchedule (\$B.51): Homes 12%, Services 25%.

T.5.1 Guest-facing total

LINE	CALCULATION	AMOUNT
Accommodation (3 nights × AED 800)	net subtotal	AED 2,400.00
Cleaning service	net	AED 200.00
Subtotal (VAT-exclusive)	—	AED 2,600.00
VAT @ 5%	2,600 × 0.05	AED 130.00
Tourism Dirham Fee	2BR Deluxe HH × 3 nights × AED 30	AED 90.00
Guest total	—	AED 2,820.00

UI displays the all-inclusive nightly rate (AED 840 = 800 + 5% VAT) and the TDF as a separate line, per UAE Consumer Protection display rules.

T.5.2 DSS retains / remits

WHAT DSS RETAINS	CALCULATION	AMOUNT	REMITTANCE
VAT on accommodation	2,400 × 0.05	AED 120.00	FTA, quarterly
VAT on cleaning	200 × 0.05	AED 10.00	FTA, quarterly
Tourism Dirham Fee	3 × 30	AED 90.00	DET, monthly
DSS commission · Homes	2,400 × 12%	AED 288.00	DSS income
VAT on DSS commission	288 × 0.05	AED 14.40	FTA, quarterly
DSS commission · Services	200 × 25%	AED 50.00	DSS income
VAT on Services commission	50 × 0.05	AED 2.50	FTA, quarterly

T.5.3 Host payout

COMPONENT	CALCULATION	AMOUNT
Net accommodation	2,400	AED 2,400.00
Less DSS commission	(288)	(AED 288.00)
Less VAT on commission	(14.40)	(AED 14.40)
Net cleaning service	200	AED 200.00
Less Services commission	(50)	(AED 50.00)
Less VAT on Services commission	(2.50)	(AED 2.50)
Net host payout	—	AED 2,245.10

Note: the host receives the net rate (VAT they collected goes to FTA via DSS, since DSS is the deemed VAT collector). The host's own VAT registration status doesn't char this flow — DSS handles VAT remittance on the supply.

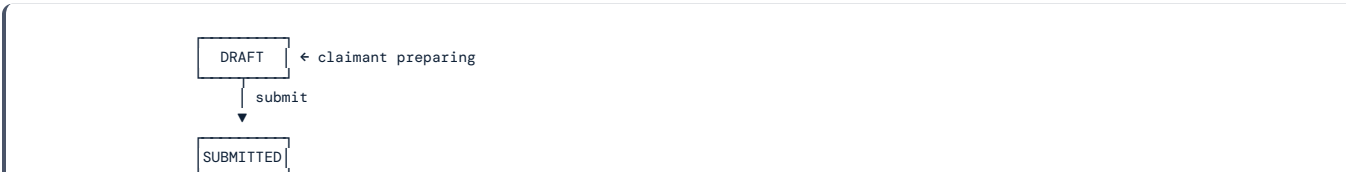
T.5.4 Reconciliation invariants

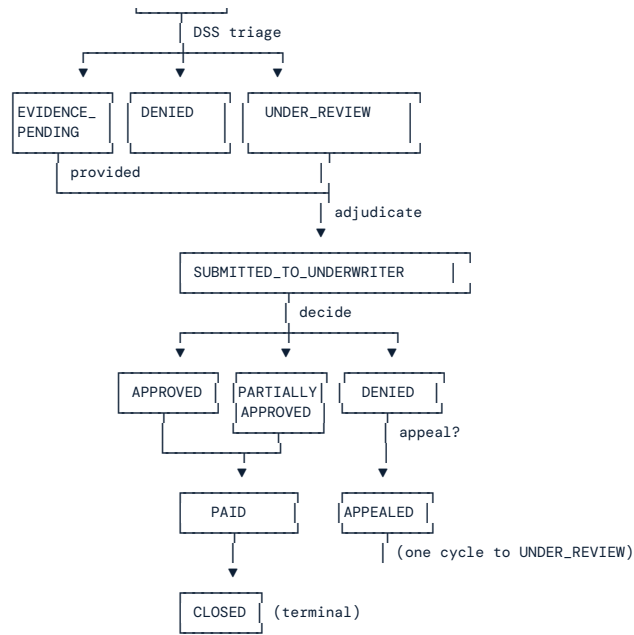
- Guest pays = Host receives + DSS commission (net) + DSS commission VAT + All supply VAT + TDF
 $2,820.00 = 2,245.10 + (288 + 50) + (14.40 + 2.50) + (120 + 10) + 90$
- VAT collected = sum of all line-level VAT (130 supply VAT + 16.90 commission VAT = AED 146.90 to FTA)
- TDF capped at 30 nights. For long-stay over 30 nights, TDF total = max_30_nights × matrix_rate (does not scale linearly past 30).
- VAT applies to all stay lengths. 90-night extended_stay carries the same 5% on accommodation as a 3-night short stay — DET classification, no duration, drives VAT treatment.

§U Claim lifecycle v7.1

⚠ **PROVISIONAL.** Lifecycle is locked; specific cover values pending E1 business confirmation.

U.1 State machine





Terminal: PAID · CLOSED · DENIED (after appeal window) · WITHDRAWN

U.2 SLAs

TRANSITION	SLA
Owner DRAFT → SUBMITTED	≤ 72h from incident
SUBMITTED → UNDER_REVIEW (DSS triage)	4h business
EVIDENCE_PENDING → SUBMITTED (owner provides)	7 business days; auto-deny on expiry
UNDER_REVIEW → SUBMITTED_TO_UNDERWRITER or DENIED	5 business days
Underwriter decision	5-15 business days (per contract)
APPROVED → PAID	14 business days
Denial → Appeal window	14 business days
Total target	< 45 calendar days end-to-end

U.3 Depreciation schedule (provisional)

For damaged items > 2 years old, depreciation applied. **Formula:** $\text{payout} = \text{claimed_amount} \times (1 - \text{depreciation_rate} \times \max(0, \text{age_years} - 2))$
 capped at `cover_per_incident_aed`.

ITEM CATEGORY	ANNUAL DEPRECIATION
Electronics (TVs, audio)	20%
Appliances (fridge, washing machine)	12%
Furniture (sofas, beds)	8%

Soft furnishings (linen, curtains)	25%
Fixtures (light fittings, taps)	5%
Structural	3%
Decor (paint, wallpaper)	15%
Kitchenware	20%

§1

Executive summary

v7.0

Vision

A vertically integrated short-stay platform for the GCC, comprising three product lines sharing one operating system:

1. **Marketplace** — curated demand engine (DubaiShortStay.com, 12% commission, Plum-Guide-tier curation)
2. **PMS** — property operations cockpit (calendar, pricing, finance, housekeeping, maintenance)
3. **Channel Hub** — OTA distribution + reservation sync (Airbnb, Booking.com, Vrbo, Expedia, Agoda)

Year-3 KPI targets

KPI	YEAR 1	YEAR 3
Live properties	200	5,000
Bookings / month	800	30,000
OTA-sourced bookings	30%	60%
API requests / month	500K	50M
Cities live	1 (Dubai)	6 (Dubai, Abu Dhabi, Riyadh, Jeddah, Doha, Valencia)
ARR	AED 4M	AED 95M

§2 Platform shape & evolution

v7.0

PHASE	MONTHS	FOCUS	ENGINEERING
0	0	Prototype	1 person
1 — MVP	1–6	Marketplace launch, 50 live properties	5
2 — PMS	6–12	Internalise ops, decommission legacy WordPress	10
3 — Channel Hub	12–18	OTA distribution, SaaS billing, self-served onboarding	16
4 — Enterprise	18–24	GCC expansion (KSA / Bahrain / Qatar) + Spain market + AI ops · multi-tenancy descoped v8.3	25
5 — Mobile + AI	24+	Native apps + AI pricing/messaging/forecasting	40+

△ **Critical principle:** don't build Phase 3 architecture in Phase 1. Build Phase 1 with clean enough boundaries that Phase 2–3 can refactor incrementally. Start with the "majestic monolith"; split into microservices only when service boundaries hurt.

§3 Single-platform model REWRITTEN v8.3

△ **v8.3 — multi-tenancy removed.** Previous design assumed a SaaS model where DSS could be sold as white-labeled platform to other operators (each their own tenant with isolated schema). v8.3 drops this assumption. DSS is a single marketplace operated by Keys Please Holiday Homes Rental Dubai LLC. There is no concept of multiple tenants. Hosts are not tenants — they are accounts within the single platform.

The data model is:

```
DSS Platform (singular)
├── Platform staff    - DSS team operating the marketplace (RBAC-controlled)
├── Hosts            - entities listing properties (DET-licensed, like Keys Please anchor + Premier Stays, Hub Living, etc.)
├── Guests           - consumers booking stays
├── Providers        - Experience and Service providers

PostgreSQL schema:
├── public           - shared reference data (cities, amenities, DET classes, ...)
└── dss              - all operational data (no per-tenant schemas)
```

Data scoping: Resources scope to `host_id` (for host-owned data like properties, payouts) rather than `tenant_id`. RBAC at the application layer enforces that hosts can only see/edit their own data. No row-level security policies are required at MVP; if needed in future, RLS would scope by `host_id`.

JWT claims: `user_id`, `role` (platform role enum), and — for host-side users — `host_id`. The previously-planned `X-Tenant-Id` header is removed.

If multi-tenancy is ever revived (Phase 4+ for Spain/GCC white-label or sovereign-cloud deals), it would be re-architected as a distinct deployment scheme — not folded back into the DSS marketplace SRS.

§4 Roles & RBAC SIMPLIFIED v8.3

Platform roles · 13 internal (DSS staff)

Super Admin · Platform Admin · Engineer/SRE · Finance Admin · Trust & Safety Admin · Ops Manager · Account Manager · Customer Support L1 · Customer Support L2 · Curation · Content/Marketing · Read-only Auditor · API Partner Admin. RBAC matrix in admin · Roles & access module.

Host roles · 3 per host org (simplified v8.3)

HOST_OWNER · full access to own host org's properties, bookings, payouts, payment methods, KYC.

HOST_STAFF · manage own host org's properties + view bookings; no payout or financial access.

HOST_VIEWER · read-only access to host's own data (for auditors, accountants).

v8.3 simplification: previous design carried 8 tenant-side roles (Owner, Property Manager, Front-Desk, Housekeeping Lead, Housekeeper, Maintenance Tech, Finance View Auditor). Most reflected PMS-style operations DSS no longer scopes (Housekeeping + Maintenance decommissioned v7.5; multi-tenant SaaS removed v8.3). Hosts can't have simple roles internally; their broader staff structure is their concern, not DSS's.

Audit log schema (every privileged action)

```
CREATE TABLE audit_log (
  id BIGSERIAL PRIMARY KEY,
  occurred_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
  host_id UUID,           -- nullable; null = platform-staff action
  actor_user_id UUID NOT NULL,
  actor_role VARCHAR(40) NOT NULL,
  actor_ip INET,
  resource_type VARCHAR(40) NOT NULL,
  resource_id UUID NOT NULL,
  action VARCHAR(40) NOT NULL,
  before JSONB,
  after JSONB,
  reason TEXT,
  request_id UUID NOT NULL
);
```

Retention: 7 years for financial / regulatory; 2 years for general.

§5 Functional surfaces v7.0

1. **Guest marketplace** — covered by v6.x
2. **Host / PM cockpit (PMS)** — multi-property dashboard, calendar, pricing manager, reservations, owner statements, housekeeping, maintenance, st tasks
3. **Operations module** — DSS-internal (booking queue, refund auth, KYC review, curation, review moderation, SLA alerts, fraud signals)
4. **Finance module** — payout reconciliation, VAT register, TDF register, withholding, chargebacks, owner statements
5. **Channel Manager** — detailed in §7

§6 API architecture v7.0

Design principles

1. REST primary, GraphQL secondary
2. Versioned in URL (`/api/v1/...`)
3. Resource-oriented, plural nouns
4. Idempotency keys on writes (`Idempotency-Key` header)
5. Standard error envelope
6. Webhooks for async (HMAC-SHA256 signed)
7. Rate limited per principal

Authentication

SURFACE	AUTH
Guest web/mobile	OAuth2 password / SSO → JWT (15 min access + 30 day refresh)
Cockpit (owner/staff)	OAuth2 password + mandatory 2FA → JWT
Partner API	API key + HMAC signing OR OAuth2 client_credentials
OTA → DSS webhooks	HMAC in <code>X-DSS-Signature</code> header

Standard error envelope

```
{
  "error": {
    "code": "BOOKING_DATES_UNAVAILABLE",
    "message": "Requested dates conflict with an existing reservation"
  }
}
```

```
    "message": "requested dates conflict with an existing reservation",
    "details": { "blocked_dates": [ "2026-06-18", "2026-06-19" ] },
    "request_id": "req_01HXXXX",
    "documentation_url": "https://docs.dubaishortstay.com/errors/BOOKING_DATES_UNAVAILABLE"
  }
}
```

Rate limits

PRINCIPAL	LIMIT
Anonymous (IP)	60 req/min
Authenticated user	600 req/min
Partner API · Starter	1,000 req/min · 50K/day
Partner API · Growth	5,000 req/min · 500K/day
Partner API · Enterprise	Negotiated

§7 OTA integration v7.0

🔒 Three locked rules:

1. **Never double-book.** Calendar truth lives in one place (DSS); OTAs are followers.
2. **Earliest-committed booking wins on conflict.** Loser auto-cancelled by DSS with 100% refund + DSS Coverage + 15% voucher.
3. **Be observably idempotent.** Every sync action emits an event; replaying events is safe.

Per-OTA connectivity

OTA	CONNECTIVITY	PHASE
Airbnb	API (Channel Provider tier, certification required)	Phase 3
Booking.com	XML API (OXI / OTA Connectivity)	Phase 3 (first)
Vrbo	HomeAway Partner Network → Expedia Partner Solutions	Phase 3
Expedia	EPS API (REST)	Phase 4
Agoda	YCS XML API (polling-heavy)	Phase 4
GDS (Sabre, Amadeus)	For corporate travel	Phase 5

Double-booking prevention

1. Distributed lock per property×date-range in Redis (TTL 30s)
2. Optimistic concurrency on the `availability` table (version increment)
3. OTA-side calendar push debounce (250ms)
4. Reconciliation every 15 min, diff DSS state on each channel

4. Reconciliation cron every 15 min — diff DSS state vs each channel
5. Conflict resolution: earliest `confirmed_at` wins; loser auto-cancelled

§8 Sync engine v7.0

Recommendation: Apache Kafka (AWS MSK). Need ordering per partition key, replay for backfill, throughput > 10K events/sec, persistent log for audit

Per outbound sync queue, retry strategy:

ATTEMPT	DELAY
1	Immediate
2	30 sec
3	2 min
4	10 min
5	30 min (final)
→ DLQ	Alert ops, manual intervention

§9 Database v7.0

PostgreSQL 16+ on AWS RDS / Aurora. ACID + JSONB + RLS + PostGIS + pg_partman.

Schema principles

- UUIDs (v7 sortable) as primary keys, not auto-increments
- All money in minor units (fils), `int NOT NULL`
- Timestamps always with timezone (`TIMESTAMPTZ`)
- Soft delete via `deleted_at` where domain-meaningful
- JSONB sparingly (channel mappings, pricing rule trees)

Scaling plan

MVP: single instance + read replicas (multi-AZ). Phase 3: vertical scaling on Aurora. Phase 4+: horizontal shard by `host_id` or `region_id` if scaling demands it (Citus or equivalent).

§10 Microservices v7.0

Not from day 1. Start as a well-bounded modular monolith; extract services where boundaries hurt.

SERVICE	TECH	EXTRACT WHEN
<code>booking-svc</code>	Node/TS or Go	Phase 2
<code>inventory-svc</code>	Go (low latency)	Phase 2 (high throughput)
<code>pricing-svc</code>	Python (numerical)	Phase 2
<code>integration-svc</code> (channel mgr)	Go or Node	Phase 3 (3+ channels)
<code>pms-svc</code>	Node/TS	Phase 3
<code>payment-svc</code>	Node/TS	Phase 2 (PCI scope isolation)
<code>messaging-svc</code>	Node/TS	Phase 3
<code>analytics-svc</code>	Python	Phase 4

§11 DevOps & infrastructure v7.0

Cloud: AWS, primary region `me-central-1` (UAE) — data residency for DET / PDPL comfort.

Compute: Amazon EKS (Kubernetes), 3+ node groups (general / spot / GPU). Lambda for image processing + scheduled batch.

Data plane: Aurora PostgreSQL · ElastiCache Redis · MSK (Kafka) · SQS · S3 + CloudFront · OpenSearch · ClickHouse Cloud · KMS · Cognito.

CI/CD: GitHub Actions → ArgoCD (GitOps) → EKS. Canary deploys (5% → 50% → 100%) with auto-rollback on SLO breach.

IaC: Terraform. State in S3 (encrypted, versioned).

Observability: Prometheus + Grafana + OpenTelemetry + X-Ray + Sentry + CloudWatch.

§12 Security & compliance v7.0

REGIME	SCOPE	WHEN
UAE PDPL (Federal Decree-Law 45/2021)	All UAE data subjects	Day 1
DET (Dubai)	Holiday Homes Permit, TDF	Compliant via Keys Please
FTA (UAE Federal)	VAT registration, quarterly returns	Day 1
PCI DSS SAQ-A	Card handling (tokenised, no PAN)	Day 1
GDPR	EU guests (Valencia)	Phase 4
PCI DSS Level 1	If volume > 6M transactions/year	Phase 4
SOC 2 Type II	Enterprise customers (corporate accounts) will demand it	Phase 3 (start posture work Phase 2)
ISO 27001	Enterprise / government deals	Phase 4

§13 Performance & SLOs v7.0

SURFACE	TARGET P95
Marketplace home/search	< 600 ms
Booking creation	< 1,500 ms
Property detail page	< 800 ms
API GET endpoints	< 200 ms
API write endpoints	< 600 ms
OTA sync (outbound)	< 5 sec
OTA webhook ingest	< 100 ms

Uptime (marketplace + API)	99.95%
RPO	5 min
RTO	1 hour

Peak concurrency target: 500 concurrent booking attempts (DSF, Eid, NYE).

§14

Third-party integrations

v7.0

CATEGORY	MVP	PHASE 2+
Payments	Stripe · Tabby · Bank transfer	Checkout.com · Tamara · Ziina
Comms	Twilio · SendGrid	WhatsApp Business API · FCM (mobile push)
Maps	Google Maps Platform	Mapbox (cost optimisation)
KYC	(Manual)	Onfido / Veriff · Sumsub
Pricing	—	PriceLabs · AirDNA
CRM	HubSpot	Salesforce · Intercom · Zendesk
Analytics	GA4 · GTM	Mixpanel · Hotjar · Segment

§15

AI & automation roadmap

v7.0

Sequenced by ROI, not by tech excitement.

Phase 2 (clear ROI)

1. AI pricing assistant (PriceLabs integration + DSS proprietary signals)
2. Smart guest message draft (saves 60–70% of host response time)
3. Booking fraud detection (risk score every booking)
4. Photo quality scoring (pre-curation filter)

Phase 3–4

5. Occupancy forecasting (90-day per-property)
6. Predictive maintenance
7. AI guest support (WhatsApp first-line)
8. Listing copy generator (EN + AR)
9. Review summarisation + theme extraction
10. Dynamic badge assignment (auto-award Top badge if UQS ≥85, auto-award Dubai Favorite if rating ≥4.8)

§16 SEO & growth v7.0

Programmatic SEO at scale. Targets:

PATTERN	Y1 / Y3
City	1 / 6
Area	12 / 60
Area × bedrooms	60 / 360
Area × amenity	100 / 600
Building	30 / 200
Property	200 / 5,000

§17 MVP vs Enterprise roadmap v7.0

Phase 1 MVP (Months 1-6)

Must-have: marketplace web · property detail + search + area pages · 3-step booking flow · Stripe Card + Bank + Tabby · owner portal (basic) · admin console · DET permit handling · TDF + VAT per \$T.

Defer: channel manager (now Phase 1 per ADR-001), mobile apps, AI deep features, Arabic UX. (Multi-tenant SaaS removed entirely v8.3.)

Phase 2 (Months 6-12)

Full PMS modules · dynamic pricing · WhatsApp · AI message drafting · fraud scoring · multi-currency · multi-locale (EN + AR) · mobile-responsive coc

Phase 3 (Months 12-18)

Channel manager deepening (per ADR-001) · partner API · SOC 2 posture · Spanish UX (Valencia readiness). *Self-served tenant signup / subscription / multi-tenant data isolation removed from roadmap v8.3.*

Phase 4 (Months 18-24)

Native mobile apps · AI ops copilot · predictive maintenance · GCC expansion · corporate B2B pipeline · white-label · ISO 27001.

Phase 5 (Months 24+)

GDS connectivity · AI listing copy + photo curation · crypto · government deals.

§18 Risks & mitigations v7.0

RISK	LIKELIHOOD	IMPACT	MITIGATION
OTA certification delays	High	High	Apply early, start with Booking.com (faster), accept manual-sync interim
Double-booking under load	Medium	Catastrophic	Distributed locks, optimistic concurrency, reconciliation cron, DSS Coverage absorbs financial impact
OTA API breaking changes	High	Medium	Versioned adapters, contract tests, paid Connectivity Partner status
Payment provider outage	Medium	High	Multi-provider (Stripe + Checkout.com fallback)

Regulatory change (DET / VAT / KYC)	Medium	High	Modular tax engine (rates configurable, not hardcoded)
Single-region outage	Low	Catastrophic	Cross-region read replica + quarterly DR drills
Data breach / PII exposure	Medium	Catastrophic	Encryption at rest, least-privilege, pentest, bug bounty, cyber insurance
Owner churn to Guesty / Hostaway	Medium	High	Bundle (marketplace + PMS + channels) makes leaving costly
Marketplace cold start	High	Existential	Bootstrap with Keys Please's existing 200+ properties, broker partnerships
Engineering capacity	High	High	Phase strictly; managed services (Cognito, Stripe Billing) wherever possible

PART V

Future *outlines*

Phase 5+ scope. Outline-level only — detailed spec would be premature. Activate when trigger conditions are met.

§7.2 Native mobile architecture v7.2

⌚ **Trigger conditions:** MAU > 100K web users · mobile web LCP plateau · > 40% guest survey demand for native · competitor mobile advantage · investo signal. Until then: **mobile web is the strategy.**

Tech recommendation

React Native (managed by Expo) with native iOS/Android bridges for performance-critical paths.

CRITERION	REACT NATIVE	NATIVE SWIFT/KOTLIN
Code reuse w/ web	~60%	0%
Time to MVP	~3 months	~6 months
UAE talent availability	High	Moderate
For DSS	✓ Yes	Only for specific features

MVP mobile scope (v7.2.1 — Phase 5 launch)

- Search, browse, property detail, booking flow
- Account management + booking history + active stay info
- In-app messaging
- Push notifications (booking events) via FCM + APNS
- Apple Pay / Google Pay
- Biometric login

Notification architecture

Single `notification.dispatched` event drives email/SMS/push/in-app. Channel preferences honoured. Push priority categories (CRITICAL booking

events, HIGH SLA, LOW marketing). Quiet hours 22:00–08:00 local for non-CRITICAL.

Deep linking

iOS Universal Links + Android App Links. URL → screen map locked for: `/property/:id`, `/area/:slug`, `/booking/:reference`, `/host/dashboard`, `/search?...` .

Offline support (MVP)

Read-only fallback. Last-viewed property cached 24h. Active booking + check-in code + WiFi info always offline. Wishlists cached 7 days. Search results NOT cached.

Performance targets

METRIC	TARGET
Cold start (warm storage)	< 2.5s
Search results TTFP	< 1.2s
Booking flow transitions	< 400ms
Bundle size (initial)	< 8 MB

§7.3 Sovereign cloud variant v7.3

- **Trigger conditions:** UAE federal/emirate RFP requires sovereign · regulated enterprise contract · sovereign wealth fund / holding company deal · GCC country requires in-country hosting. Until then: AWS me-central-1 sufficient.

Architecture variant

For triggering deals: **Core42 / G42 Cloud** (UAE-sovereign, Azure-based) as primary sovereign provider. One sovereign *deployment* per deal — separate instance, no shared infrastructure with commercial DSS. (Previously framed as a tenant; v8.3 reframes as separate deployment.)

Required certifications

CERTIFICATION	REQUIRED FOR
NESA IA Standards	Federal / critical infrastructure
ISO 27001	Most enterprise / government RFPs
SOC 2 Type II	Enterprise SaaS
CSA STAR Level 2	Cloud-specific reassurance
DESC (Dubai)	Dubai government deals
Cybersecurity Council (CSC)	Critical sector entities

Posture work timeline: 9–18 months from trigger to certification. Cannot compress below 9 months.

Sovereign-specific changes

- **Identity:** UAE Pass federation + YubiKey for admin access
- **Data residency:** all data physically in UAE; backups to second UAE DC (no cross-region)
- **Encryption:** Bring-Your-Own-Key minimum; Hold-Your-Own-Key preferred (government HSM)
- **Air-gapped option:** no internet egress; offline patch bundles; sovereign equivalents for Stripe/Twilio/etc.

Deployment models offered

1. **Dedicated sovereign deployment** — DSS-managed on G42 (separate instance)
2. **Customer-hosted deployment** — licensed software, customer ops
3. **Air-gapped customer-hosted** — classified environments

Commercial model

Enterprise-tier custom: onboarding fee AED 500K–2M one-time · annual licence AED 1M–10M · 24/7 named SRE support · source code escrow typically required.

Pre-trigger preparation (cheap now)

Current AWS me-central-1 architecture already aligns. Document data flows. Track third-party dependencies for substitution paths. SOC 2 Type II preparation work in Phase 2–3 (foundational). Auth-svc designed with pluggable providers (Cognito ↔ UAE Pass swap). Quarterly sovereign-readiness review.

PART VI

Open *items*

Strategic decisions pending. Entity gaps identified by audit. Backlog evolves with each version bump.

§2.1 Open strategic decisions

From Design Review §9. v7.0 resolved 7 of 8; v7.1 moved E1 from BLOCKED to PROVISIONAL.

#	QUESTION	STATUS
A1	Positioning voice (luxury vs monthly)	✓ Resolved — "curated, regional GCC"
A3	Luxury tier benchmark	✓ Resolved — Plum Guide curation + Guesty PMS depth + GCC-native
B1	Launch-day inventory count	✓ Locked — 50 by Month 6
B3	Instant Book vs RTB	✓ Default Instant Book; per-property override
C1	Homes commission %	⚠ Default 12% (locked); final policy lock pending
C2	Payment methods at MVP	✓ Resolved — Stripe Card + Bank + Tabby
D1	Valencia scope	✓ Resolved — Channel Hub SaaS Phase 3, full ops Phase 4
E1	DSS Coverage policy details	🟡 PROVISIONAL — schema locked in v7.1, specific values pending business

⚠ **E1 — what's still needed:** Underwriter selection (recommend RFP to AIG, Allianz, Tokio Marine, Oman Insurance) · Cover amount per incident (provisio AED 100,000) · Premium absorption policy · Exclusion list legal review · Claim escalation path. Schema is locked; final values swap in via config when ready.

§Z.2 Entity backlog

Status updated after v6.4.1, v7.0.1, v7.0.2, v7.1, v7.4, v7.5. All entities now resolved; only E1 (DSS Coverage policy values) remains provisional pending business confirmation.

ENTITY	STATUS	RESOLVED IN
Insurance / DSS Coverage Policy	🟡 Provisional	v7.1 (schema locked; values pending E1)
Claim	🟡 Provisional	v7.1 (schema locked; values pending E1)
Wishlist · WishlistItem	✓ Resolved	v7.0.1
PromoCode · Discount	✓ Resolved	v7.0.1
RatePlan	✓ Resolved	v7.0.1 + extended in v7.4
PricingRule	✓ Resolved	v7.0.1
EmailTemplate	✓ Resolved	v7.0.1
BookingModification	✓ Resolved	v6.4.1
DisputeCase	✓ Resolved	v6.4.1 (extended in v7.5 to absorb maintenance complaints)
PayoutBatch · Payout	✓ Resolved	v6.4.1
HousekeepingTask	🔕 Decommissioned	v7.5 — operator concern; replaced by Service §B.2 + Review §B.14 + DisputeCase §B.16
MaintenanceTicket	🔕 Decommissioned	v7.5 — operator concern; replaced by Communications §S + DisputeCase §B.16 + Coverage §E
Vendor	🔕 Decommissioned	v7.5 — host's commercial relationship; Provider §B.3 remains for Experience/Service marketplace
CorporateAccount	✓ Resolved	v7.0.2 + extended in v7.4
UtilityCharge	✗ Decommissioned v8.2	Removed — utilities are a host operating cost. Rates all-inclusive.
SecurityDeposit	✓ Resolved	v7.4
ShortStayAgreement	Decommissioned v8.3	Long-stay scope removed — booking record is the contract for short stays
RelocationPackage (reframed)	✓ Resolved	v7.4 + reframed v7.5 as marketplace Service bundle
DepositInspection (renamed)	✓ Resolved	v7.4 + renamed/narrowed v7.5 — host+guest joint, marketplace-neutral, move_in/move_out o

Total entities locked: 33. (Was 36 after v7.4; v7.5 decommissioned 3 operator-side entities.) See §B.0 through §B.36 across the Foundations section, §B.26–28 retained as decommission markers for engineering reference.

§Z.3 Architecture decisions (ADRs)

All 14 ADRs locked. Engineering may begin development. See `DSS-ADR-Compendium.md` + `DSS-ADR-001-Channel-Manager.md` for full details.

ADR	DECISION	REVERSIBILITY
001	Channel Manager: Integrate Channex (primary), Hostaway (fallback)	Medium
002	Backend framework: Node.js + TypeScript + Fastify; Next.js for SEO	Medium
003	Database: PostgreSQL 16 on AWS RDS, primary + read replica from day 1	Low
004	Multi-tenancy: RLS in db core at MVP; schema-per-tenant from Phase 2 — Superseded v8.3: single-platform model adopted, no multi-tenancy. See §3.	Decommission v8.3
005	Queue: SQS at MVP, MSK Kafka from Phase 2	High
006	Payments: Stripe + Tabby at MVP; Checkout.com fallback Phase 2	Medium
007	Search: Postgres tsvector at MVP; OpenSearch from Phase 2	High
008	Observability: Sentry + CloudWatch + OpenTelemetry; Datadog Phase 2	High
009	Auth: AWS Cognito with custom UI	Low

O09	Auth: AWS Cognito with custom UI	Low
O10	IaC: Terraform with S3 remote state	Medium
O11	Deployment: ECS Fargate at MVP; EKS from Phase 2	High
O12	Mobile: Defer per v7.2; React Native + Expo when triggered	High
O13	AI: PriceLabs Phase 2; LLM guest support Phase 3; in-house ML Phase 4	High
O14	Region: me-central-1 primary; eu-central-1 for Spain Phase 4	Low

§Z.4 Suggested next bumps

VERSION	SCOPE	STATUS
v6.4.1	BookingModification + DisputeCase + PayoutBatch + refund edge cases	✓ Done
v7.0.1	Wishlist + PromoCode + RatePlan + PricingRule + EmailTemplate	✓ Done
v7.0.2	HousekeepingTask + MaintenanceTicket + Vendor + CorporateAccount	✓ Done
v7.1	Insurance + Claim entities (PROVISIONAL)	✓ Done — pending E1 confirmation
v7.4	Long-stay + corporate workflows (UtilityCharge + SecurityDeposit + LeaseAgreement + RelocationPackage + InspectionReport)	✓ Done
v7.1.1	Final policy values (cover amount, underwriter, exclusions)	Awaits E1 business decision (Insurance Underwriter RFP issued)
v7.4.1	Refinements after first 5 lease agreements signed in production	Triggered by first signed leases
v7.2	Native mobile apps — detailed component spec	Outline only; activate on Phase 5 trigger
v7.3	Sovereign cloud variant — detailed deployment spec	Outline only; activate on government deal trigger

SRS ADDENDUM · APPROVED 2026-06-11

v9.1 — Guest Experience Completeness (ALL Scopes A–I approved)

All nine scopes item-by-item approved by HM on 2026-06-11. Rule-based + curated only; no ML. No third-party cookies.

§v9.1-0 · Foundational Decisions (Confirmed)

DECISION	APPROVED CHOICE
Visitor ID generation	Random UUID v4 + persistent cookie (1-year TTL). International users, not GCC-only.
Event log retention	Hot delete after 90 days. No archive.
Jurisdiction detection	IP geolocation (automatic).
Editorial curation	New <code>content_manager</code> admin role: content:create / edit / publish / archive.
Crypto payments	Excluded from v9.1 (defer to v9.2+).
Pre-arrival comms	Automated workflow, triggered 24h before check-in.

§AC · Trust & Credibility (Scope A — APPROVED)

- 🏆 **Verified Host badge** — visual emoji badge. Criteria: Rating ≥ 4.8 AND response time $< 2h$. Shown on cards + listing detail.
- **Zero-review empty state** — listings with no reviews show: "Host verified • Responds in $< 2h$ • Member since [year]" instead of a blank section.
- **Reviews first-class** — prominent dedicated section on listing detail. Display: average rating only (🌟4.8) — no sentiment breakdown.
- **Site-wide trust block: FOOTER** — admin-managed via new entity **§B.59 FooterContent** (licence number, company info, support email/hours, payment marks, SSL, links). Editable by footer manager from admin.
- **Cancellation rate display** — 📌 DEFERRED to v9.2 (still feeds Top Provider badge calc in backend; not displayed).

§AB · Price Transparency (Scope B — APPROVED)

- **Search cards**: "AED 850/night • Total: AED 5,950 (7 nights)" — total contextual to selected dates.
- **Listing detail price modal** (tap price / "View price details"):
 - Dates summary + **"Change dates" button**
 - Nights $\times$ rate line
 - Discount line (green, if applicable)
 - **"Price for stay"** total (prominent)
 - Free-cancellation note
 - "Proceed to checkout" CTA
- **Excluded from modal**: cleaning fee, DSS service fee, subtotal, VAT lines (removed per approval). Fee/VAT detail appears only at checkout payment step.

§AA · Booking Funnel & Checkout (Scope C — APPROVED)

9-step funnel (Airbnb-pattern, account gated before payment):

1. Review (dates, guests, price) — Change buttons inline
2. Login or sign up (email/phone or Google/Apple social)
3. Create account (name, DOB, email, phone)
4. Community Commitment (agree to values screen)
5. Payment (add method + charge)
6. **Confirmation** ✅ **"Booking confirmed"**
7. Verify email (link in inbox, valid 7 days)
8. Verify phone — user chooses **SMS OTP or WhatsApp OTP** (code valid 15 min)
9. Upload passport copy (KYC L1, deadline 24h post-booking)

- **Steps 7–9 run AFTER confirmation** — never gate the booking.
- **Progress indicator:** "Step N of 9" on every funnel screen.
- **Trip context persistence:** dates/guests/location persist across funnel and across all three lines (Stays/Experiences/Services) via URL state (§K) + session.
- **Payment methods (v9.1):** Stripe (card), Amazon Pay (card), Tabby (BNPL), Apple Pay, Google Pay. Bank transfer dropped; crypto deferred to v9.2.
- **KYC ladder:** LO = account created → L1 = email + phone verified + passport uploaded (within 24h) → L2 per existing §B.13.

§AD · Mobile-First & Performance (Scope D — APPROVED)

- **Skeleton strategy: MINIMAL (Option A)** — first paint ~0.8s renders nav + hero + search bar only. All other modules (featured collections, reviews, rails, footer) async lazy-load below the fold with skeleton placeholders. Heavy/personalized modules never block first paint.
- **Date picker: CALENDAR GRID (Option B)** — familiar month grid, with enlarged date cells (40px touch targets) for thumb accuracy.
- **Guest picker:** +/- stepper buttons, 36px minimum touch targets.

§B.53–B.59 · New Entities (v9.1)

ENTITY	STATUS	PURPOSE
B.53 VisitorIdentity	✓ Approved (decisions locked)	UUID v4 + 1-yr cookie; stitched to account on login; no third-party cookies.
B.54 ConsentRecord	✓ Approved (decisions locked)	Consent architecture; jurisdiction via IP geolocation; international coverage.
B.55 EventLog	✓ Approved (decisions locked)	Search/view/save/funnel events + trip context; hot-delete at 90 days.
B.56 EditorialContent	✓ Approved (Scope E)	Curated collections; content_manager role approved.
B.57 RecommendationRail	✓ Approved (Scope G)	Rule-based labeled rails, dismissible.
B.58 AbandonedCheckout	✓ Approved (Scope I)	Checkout abandonment + consent-gated recovery.
B.59 FooterContent	✓ Approved (Scope A)	Admin-editable footer trust block.

§4 · RBAC Update

New role: `content_manager` — permissions: `content:create`, `content:edit`, `content:publish`, `content:archive`. Footer trust block editable via `content:edit` on FooterContent.

§Y · Editorial Content & Cold-Start (Scope E — APPROVED)

- **§B.56 EditorialContent** — collection types: area ("Living in JBR"), seasonal ("Summer escapes"), event ("F1 weekend stays"), lifestyle ("Great for families"). Each = title + hero image + filter rules that auto-populate matching listings.
- **Suggested AND managed by admin:** system may *suggest* collections (e.g., "JBR has 12 listings — create an area collection?") but `content_manager` decides, edits, publishes. No auto-publishing, ever.
- Homepage leads with editorial collections as the managed-content surface; honest curated empty-states wherever personalization would otherwise sit.

§Z · Cross-Line UX (Scope F — APPROVED)

- **Unified trip context:** dates/guests/location entered once, carried across Stays, Experiences, Services tabs (URL state §K + session).
- **Attach surface:** confirmed Home booking triggers contextual Experiences + Services scoped to booked dates + location. Appears on (1) confirmation screen and (2) automated pre-arrival email (24h before check-in).

§X · Recommendation Rails (Scope G — APPROVED)

- **§B.57 RecommendationRail** — rule-based only, NO ML. Rail types: because_searched, available_dates, popular_in_area, editorial_pick.
- **Every rail labeled with reason** ("Because you searched JBR") + feasibility sub-label ("All available Jun 19–26 · fits 2 guests").
- **Hard feasibility constraints:** no out-of-date, over-capacity, or out-of-budget results. Every card bookable now.
- **Dismissible with Undo;** dismissals logged as negative signal in EventLog (respects 90-day TTL).
- Same rail engine powers §Z attach surface (one component, two use cases).

§H + §H-1 · Identity, Tracking & Consent (Scope H — APPROVED)

- **§H VisitorIdentity (B.53):** UUID v4 + persistent first-party cookie (1-yr TTL), stitched to account on login. **NO third-party cookies anywhere, ever.**
- **§H ConsentRecord (B.54):** consent modal on first visit; jurisdiction via IP geolocation; international coverage (not GCC-only); gates analytics, recovery, personalization.
- **§H-1 EventLog (B.55):** search/view/save/funnel/book/rail_dismiss events with trip context. Hot delete at 90 days. Live session intent overrides history.

§AE · Recovery & Re-engagement (Scope I — APPROVED)

- **Abandoned-checkout (§B.58):** cart saved 7 days when guest leaves payment step. Email (+ SMS if phone provided): listing, dates, price recap + recovery link restoring checkout state exactly where they left off.
- **Abandoned-search:** trigger = 5+ searches, listings viewed, no booking, session ended. One email: "Still looking in [area]? 3 homes available for your dates" with restore-search link.
- **Consent-gated:** both flows require `email_recovery` consent = true (§B.54). No consent = no message, ever.

v9.1 Final Status — ALL SCOPES APPROVED

§15.ML · ML Scope Fence (v9.1)

ML is permitted **exclusively for admin-facing decision support** (MORCEL pricing suggestions, §15). ML shall **never** influence guest-facing search ranking, recommendation rails (§X), badge eligibility (§B.O–B.2), or editorial content (§Y). Every accepted MORCEL suggestion is recorded in the audit log. Host-facing rate suggestions (Phase 2) require explicit admin enablement. Guest-facing guarantee: *nothing a guest sees is decided by a black box.*

§HP · DSS Host Portal (LOCKED 2026-06-11)

Canonical file: `DSS-Host.html` (renamed from DSS-Host-Master; design and structure LOCKED — changes extend panes, never restructure).

Locked panes (12):

1. **Dashboard** — portfolio overview, pending actions
2. **Manage listings** — per-property details, rates, policies

3. **Bookings** — all bookings; Monthly / Weekly / In-house filters; CSV export
4. **Rate calendar** — season bands (e.g. GITEK, Peak, Summer), 12-month view, custom overrides
5. **Extensions** — extension requests, 22h response window, approve triggers payment link
6. **Earnings** — MTD/YTD revenue, next D+1 payout
7. **Payouts** — D+1 wire transfers, IBAN management
8. **Invoices** — PDF per booking, monthly ZIP download
9. **Messages** — guest + DSS support comms
10. **Reviews** — blind system, 14-day window
11. **Profile & KYC** — completion meter, Emirates ID verification
12. **Channel distribution** — admin-gated OTA distribution (§B.52)

§B.10/§B.o · Stay Duration Rules (v9.2-C1 — APPROVED)

- **Platform: duration-agnostic.** No platform-level maximum stay. Booked-nights distribution tracked as analytics metric while market traction is analyzed.
- **Host-level controls:** Property, `min_nights` (default 1) and Property, `max_nights` (default null = unlimited). Editable anytime in DSS Host; applies to new bookings only.
- **Admin:** observes distribution, flags outliers; does not override host choice. The v8.3 admin default of 27 nights is retired.
- **Glossary:** "All bookings are short-stay" replaced with "DSS is duration-agnostic; stay length is a host-level setting."
- No stay classifications reinstated; no legal position taken on 90+ nights (gated on counsel sign-off).

§B.4 · RatePlan — Host Pricing Engine (v9.2-C2 — APPROVED)

- **Three base rates per listing:** `rate_nightly` · `rate_weekly` · `rate_monthly` — each host-set directly, never auto-derived. Applies equally to SR and HH.
- **LOS discount tiers:** host-defined breakpoints (suggested defaults 7+/30+/90+/180+ nights, fully editable — add/remove/change any tier) with % off base. Guest-facing label: "Stay 30+ nights — save 18%".
- **Pricing precedence (locked):** base rate → LOS tier discount → seasonal Rate-calendar override. The locked Rate calendar pane is the seasonal layer.
- **Stay rules co-located:** `min_nights` / `max_nights` edited in the same pricing panel (C1).
- **Live preview:** "A 90-night guest pays AED X/month" sample shown before save.
- **All-inclusive (§A) unchanged** — no utility surcharges at any duration.

§B.6o · PaymentSchedule — Long-Stay Payment Security (v9.2-C3 — APPROVED)

- **Monthly billing:** bookings >35 nights bill in monthly cycles. Month 1 at confirmation; subsequent months charged 7 days before each period. Full schedule visible to guest and host.
- **Charge trigger — OPEN (gateway terms under review by HM):** Path A = MIT auto-charge (saved card, 3DS mandate at checkout, merchant-initiated cycles). Path B = payment link per cycle (sent T-7, guest approves each charge). All other mechanics identical; decision pending gateway confirmation.
- **Security layer:** one refundable security month held by DSS in escrow (not the host) at confirmation. Guest-facing copy: "Pay monthly + one refundable security month."
- **Default protocol (locked):** failed charge → retry day 2 → guest notified + pay-by-link → day 5 escrow month applied + 30-day notice issued → host paid from escrow via D+1. Audit-logged; admin "Payment risk" queue.
- **Host view:** Payouts pane schedule — secured vs upcoming months per booking.
- **Deferred:** post-dated cheques · rent-guarantee insurance (v9.3 candidate) · crypto (v9.2+).

§B.10/§S · Long-Stay Disclosure & Tooling (v9.2-C4 — APPROVED)

Principle: DSS is the platform between host and guest — it never performs operations. Hosts *declare*, the platform *displays* and *measures*.

- **Housekeeping cadence:** host-declared listing attribute for 28+ night stays (weekly / fortnightly / on-request / not included). Shown in "What's included" + booking confirmation.
- **Mid-stay inspections:** host option; platform provides scheduling with mandatory guest notice + photo log (DepositInspection pattern). Record attaches to booking as §B.16 dispute evidence.
- **Maintenance response:** host declares a per-listing commitment; platform displays declared vs measured response time. Persistent gaps feed UQS engagement + badge eligibility.
- **Inclusions disclosure:** structured long-stay fields (linen, consumables, wifi, parking) replacing free text — comparable across listings, resolvable in disputes.
- **Platform comms (§S):** automated day-7 check-in message · monthly statement per billing cycle (C3) · pre-renewal nudge (feeds C6). Mediation, not operations.
- **Net:** declared-vs-delivered becomes the long-stay trust mechanism.

§B.61 · OrganizationAccount — Corporate & Sponsor Bookings (v9.2-C5 — APPROVED)

- **§B.61 OrganizationAccount:** company name · trade licence · TRN · billing contact · payment method · member seats. One account, multiple bookers. Serves employers, relocation sponsors, medical insurers.
- **Book-for-someone-else:** roles split — *payer* (org) receives PaymentSchedule + invoices (C3); *occupant* (employee/patient) receives check-in details and completes own verification (§AA steps 7–9). KYC follows the occupant; billing follows the company.
- **VAT invoicing:** issued to org TRN, monthly, aligned to C3 cycles. PO/cost-centre field per booking.
- **Org view (v1):** active stays, renewals, monthly spend, invoice downloads — scoped view inside existing guest account system, not a new portal.
- **Host impact: none.** Normal booking + "corporate" indicator; C3 security identical.
- **Deferred to v9.3+:** negotiated corporate rates · block bookings · relocation-agency API.

§B.10 · Extensions — First-Class Flow (v9.2-C6 — APPROVED)

- **Guest-initiated:** "Extend my stay" in booking view. Re-priced from host RatePlan (C2); **LOS tier recalculates on cumulative nights** — crossing a breakpoint prices the extension at the better tier automatically.
- **Host-initiated offers (new):** host sends duration + rate (pre-filled from RatePlan, host may sweeten) on any active booking. One-tap accept in guest Messages/email → C3 PaymentSchedule extends → calendar blocks → both confirmed.
- **Conflict guard:** only available nights offerable; honest "available to extend until [date]" when blocked (§F.1).
- **Pre-renewal nudge (§S):** T-14 before checkout on stays ≥28 nights, both sides prompted; per-booking opt-out.
- **Locked preserved:** 22h host response window on guest requests (DSS Host Extensions pane).
- **KPI:** extension rate per listing + platform-wide — primary long-stay traction signal (feeds C1 demand analysis).

§D/§AB/§Y/§X · Duration-Aware Search & Discovery (v9.2-C7 — APPROVED)

- **Duration-aware pricing (§AB):** searches of 28+ nights show effective monthly rate after LOS tiers — "AED 13,838/month · 90-night rate" — nightly secondary. Under 28 nights: approved v9.1 format unchanged. Displayed price always equals the searcher's actual duration price.
- **Stay-length filter (§D):** "How long?" presets (week / month / 3 months / 6+) or exact nights; filters on host min/max (C1); sort by best monthly rate.
- **Flexible long-stay dates:** month-level start + duration searches availability windows (v1 = month-start picker).
- **Segment collections (§Y):** the eight flexible-living segments become the editorial collection set ("Relocating to Dubai", "Between

homes", "Renovating? Stay nearby", "Recovery stays near Healthcare City", "Founder-friendly monthly stays") — system-suggested, content_manager-decided per Scope E.

- **Duration rails (\$X)**: new labels on the approved rule-based engine — "Available for 3 months from August" · "Hosts welcoming 90+ night stays" · "Monthly rates under AED 15K".
- **Deferred v9.3**: commute/school/hospital proximity search layers.

v9.2 Flexible Living — Changelog (all 7 changes approved 2026-06-11)

v9.2 (June 2026) — Flexible Living. Platform becomes duration-agnostic while market traction is analyzed. C1 Stay duration rules (host min/max, no platform cap) · C2 RatePlan pricing engine (3 base rates, host-defined LOS tiers, precedence base→LOS→season) · C3 PaymentSchedule long-stay security (monthly cycles, escrow month, default protocol; charge trigger pending gateway review) · C4 Long-stay disclosure & tooling (host declares, platform displays & measures) · C5 OrganizationAccount (payer/occupant split, TRN invoicing) · C6 Extensions first-class (host offers, cumulative LOS re-pricing, T-14 nudges) · C7 Duration-aware search (monthly-rate display, stay-length filter, segment collections, duration rails). New entities: B.60 PaymentSchedule, B.61 OrganizationAccount. Principle locked: DSS is the platform between host and guest — never the operator.

DSS SRS v9.1 CANONICAL · All scopes A–I approved & applied 2026-06-11